DevOrSysAdminScripts

Laurent Frédéric Bernard François Lyaudet
https://lyaudet.eu/laurent/
Laurent.Lyaudet@gmail.com

March 18, 2026

Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

Current version: 2026-03-18
Current number of commits: 619
Current git SHA1: cf89bcaa5c4fb8a8ee3242ecec6ba00c00cf9ff6
Code lines: 11001 total lines, 10021 not empty lines, 980 empty lines.

Contents

1 Files tree **4**

2 Listing of files **8**

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl . . . 11
2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh . . . 13
2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh . . . 16
2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh . . . 18
2.5 build_and_checks_dependencies/check_URLs.libr.sh . . . 21
2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh 24
2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl . . . 41
2.8 build_and_checks_dependencies/common_options.libr.sh . . . 56
2.9 build_and_checks_dependencies/comparisons.libr.sh . . . 63
2.10 build_and_checks_dependencies/create_PDF.exec.sh . . . 67
2.11 build_and_checks_dependencies/generate_from_template.libr.sh . . 79
2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh . . . 82
2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh . 84

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh 86
2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl 95
2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl 95
2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl 96
2.18 build_and_checks_dependencies/lines_counts.libr.sh 96
2.19 build_and_checks_dependencies/lines_filters.libr.sh 100
2.20 build_and_checks_dependencies/lines_maps.libr.sh 111
2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh . . . 113
2.22 build_and_checks_dependencies/python_black_complement.libr.sh . 115
2.23 build_and_checks_dependencies/python_isort_complement.libr.sh . 116
2.24 build_and_checks_dependencies/repository_data.libr.sh 118
2.25 build_and_checks_dependencies/shell_checks_complement.libr.sh . 120
2.26 build_and_checks_dependencies/split_score.exec.php 122
2.27 build_and_checks_dependencies/split_score.libr.php 123
2.28 build_and_checks_dependencies/strings_functions.libr.sh 125
2.29 build_and_checks_dependencies/too_long_code_lines.libr.sh 152
2.30 build_and_checks_dependencies/update_common_build_and_checks.exec.sh 155
2.31 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh 157
2.32 build_and_checks_variables/.gitignore 161
2.33 build_and_checks_variables/bandit.ini 161
2.34 build_and_checks_variables/bandit_baseline.json 161
2.35 build_and_checks_variables/black.toml 162
2.36 build_and_checks_variables/DevOrSysAdminScripts.html.tpl 162
2.37 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl 174
2.38 build_and_checks_variables/files_listing.tex.tpl 180
2.39 build_and_checks_variables/files_names_listing.txt 196
2.40 build_and_checks_variables/licenses_notes.txt 199
2.41 build_and_checks_variables/phpmd_baseline.xml 203
2.42 build_and_checks_variables/pylintrc.toml 203
2.43 build_and_checks_variables/repository_data.txt 204
2.44 build_and_checks_variables/ruff.toml 207
2.45 build_and_checks_variables/shellcheck.ini 208
2.46 build_and_checks_variables/shellcheck_notes.txt 208
2.47 build_and_checks.exec.sh 210
2.48 COPYING 212
2.49 COPYING.LESSER 227
2.50 git_helpers/git_checks.libr.sh 231
2.51 git_helpers/post-commit 233
2.52 git_helpers/post_commit_renaming_comment.libr.sh 235
2.53 git_helpers/pre-commit 243

2.54 IMAP_transfert.py 244
2.55 LiensUtiles.txt 250
2.56 miscellaneous/funny.sh 251
2.57 miscellaneous/spampty_data.txt 252
2.58 README.md.tpl 252
2.59 unicode.libr.php 255
2.60 upgrade_build_and_checks.exec.sh 263
2.61 wget_sha512.libr.sh 264

1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spell_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
```

```
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   01234\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456\
│   ├── a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │   an_see_that_it_works.txt
│   ├── funny.sh
│   ├── spampty_data.txt
│   └── yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│       that_you_can_see_that_it_works.txt
```

```
├── README.html
├── README.md
├── README.md.tpl
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 74 files

[ 12K Mar 18 15:12]  ./
├── [4.0K Mar 18 15:12]  build_and_checks_dependencies/
│   ├── [3.8K Mar 10 16:37]  build_and_checks.exec.sh.tpl*
│   ├── [4.1K Mar 10 16:49]  build_md_from_printable_md.exec.sh*
│   ├── [3.2K Mar 10 16:30]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.3K Mar 18 14:50]  check_shell_scripts_indentation.libr.sh
│   ├── [3.4K Mar 10 16:30]  check_URLs.libr.sh
│   ├── [ 31K Mar 18 15:05]  common_build_and_checks.exec.sh*
│   ├── [ 26K Mar 18 15:03]  common_build_and_checks.exec.sh.tpl
│   ├── [9.1K Mar 18 14:42]  common_options.libr.sh
│   ├── [6.1K Mar 18 14:25]  comparisons.libr.sh
│   ├── [ 19K Mar 18 14:21]  create_PDF.exec.sh*
│   ├── [3.7K Mar 18 14:39]  generate_from_template.libr.sh
│   ├── [2.0K Mar  5 21:22]  get_common_text_glob_patterns.libr.sh
│   ├── [2.8K Mar 10 17:37]  grammar_and_spell_check.libr.sh
│   ├── [4.0K Mar 18 14:31]  licenses_templates/
│   │   ├── [ 13K Mar 18 14:31]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [4.8K Mar 18 15:05]  lines_counts.libr.sh
│   ├── [ 13K Mar 18 14:14]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [2.0K Mar 18 14:54]  overwrite_if_not_equal.libr.sh
│   ├── [2.3K Mar 10 16:30]  python_black_complement.libr.sh
│   ├── [2.9K Mar 10 17:36]  python_isort_complement.libr.sh
│   ├── [2.0K Mar  9 14:50]  repository_data.libr.sh
│   ├── [3.2K Mar 18 14:35]  shell_checks_complement.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
│   ├── [1.9K Feb  3 20:11]  split_score.libr.php
│   ├── [ 47K Mar 18 13:07]  strings_functions.libr.sh
│   ├── [3.8K Mar 18 14:52]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar  9 16:05]  update_common_build_and_checks.exec.sh*
│   └── [5.7K Mar 18 12:56]  update_or_check_files_names_listing.exec.sh*
├── [4.0K Mar 18 15:05]  build_and_checks.exec.sh*
├── [4.0K Mar 18 15:12]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
```

```
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [ 16K Mar  8 17:00]  DevOrSysAdminScripts.html.tpl
│   ├── [ 32K Mar 18 15:12]  DevOrSysAdminScripts.tex
│   ├── [8.9K Mar 17 14:38]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 14K Mar 17 15:58]  files_listing.tex.tpl
│   ├── [5.1K Mar 10 16:14]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [5.2K Mar  8 16:52]  licenses_notes.txt
│   ├── [1.0K Mar  5 21:36]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.5K Mar 17 12:41]  repository_data.txt
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [3.5K Mar  9 14:46]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [432K Mar 18 15:12]  DevOrSysAdminScripts.html
├── [376K Mar 18 15:12]  DevOrSysAdminScripts.pdf
├── [4.0K Mar 18 14:56]  git_helpers/
│   ├── [3.8K Mar 18 14:56]  git_checks.libr.sh
│   ├── [2.1K Mar 18 12:15]  post-commit*
│   ├── [ 14K Mar 18 11:59]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Mar 18 13:17]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar  9 15:15]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├── [   0 Mar  4 10:01]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├── [   0 Mar  4 09:13]
```

```
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── [1.3K Mar  9 15:15]  funny.sh
│   ├── [  94 Mar  3 19:56]  spampty_data.txt
│   └── [   0 Mar  4 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.0K Mar 10 16:50]  README.html
├── [4.2K Mar 10 16:50]  README.md
├── [4.2K Mar  9 15:34]  README.md.tpl
├── [ 12K Mar 18 14:02]  unicode.libr.php
├── [1.4K Mar 10 16:37]  upgrade_build_and_checks.exec.sh*
└── [1.5K Mar  8 20:25]  wget_sha512.libr.sh

6 directories, 74 files
```

2 Listing of files

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;

- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
`https://www.gnu.org/software/sed/`,
`https://savannah.gnu.org/projects/sed/`.

ShellCheck can be found at the following URLs:
`https://www.shellcheck.net/`,
`https://github.com/koalaman/shellcheck`.

GNU Wget can be found at the following URLs:
`https://www.gnu.org/software/wget/`,
`https://savannah.gnu.org/projects/wget/`.

more of util-linux can be found at the following URLs:
`https://www.kernel.org/pub/linux/utils/util-linux/`,
`https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/`,
`https://github.com/util-linux/util-linux` official?

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
```

```
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\n\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
```

```
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
```

```
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
```

```
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
```

```
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
```

```
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 LFBFL_subdir="build_and_checks_dependencies"
26 # shellcheck source=common_options.libr.sh
27 source "./${LFBFL_subdir}/common_options.libr.sh"
28 # shellcheck source=lines_filters.libr.sh
29 source "./${LFBFL_subdir}/lines_filters.libr.sh"
30
31 check_one_shell_script_indentation(){
32   declare -i LFBFL_i_file_with_error=0
33   local LFBFL_some_line
34   local LFBFL_previous_line="workeduntilsomeasshole"
35   # created a command named "workeduntilsomeasshole"
36   # and called it without neither indentation, nor parameters
37   # in a .sh script XD.
38   # Manual justification of the lines L-4 and L-3 was a serendipity.
39   # I just tried a joke and was lucky ;) :).
40   local LFBFL_line_end
41   local LFBFL_previous_line_end
42   declare -i LFBFL_i_line_length
43   declare -i LFBFL_i_line_end_length
44   declare -i LFBFL_i_line_start_length
45   local LFBFL_spacing
46   declare -i LFBFL_i_offset
47   local LFBFL_substring1
48   local LFBFL_substring2
49   declare -r LFBFL_pattern='^ *-e '
50   # This grep would have been enough without some sed commands.
51   declare -r LFBFL_s_lines=$(
52     grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"
53   )
54   if [[ -z "${LFBFL_s_lines}" ]]; then
55     return
56   fi
57
58   declare -a LFBFL_arr_lines
59   mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
60   readonly LFBFL_arr_lines
61
62   for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
63     if [[ "${LFBFL_some_line}" == "--" ]]; then
64       # When switching between code fragments, we reinit.
65       LFBFL_previous_line="workeduntilsomeasshole"
66       continue
67     fi
68     # Note how it's funny that this loop always work because we check
```

```
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length-LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length-2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset+=1));;
      :) ((LFBFL_i_offset+=1));;
      [0-9]) ((LFBFL_i_offset+=2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done
```

```

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

2.5 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
```

```
50
51    declare -Ar LFBFL_substitutions=(
52      ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
53    )
54
55    local LFBFL_pattern
56    local LFBFL_base_file_name
57    local LFBFL_substitution
58    local LFBFL_substitution2
59    local LFBFL_file_path
60    local LFBFL_s_files_paths
61    declare -a LFBFL_arr_files_paths
62    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
63      [[ LFBFL_i_verbose -eq 1 ]]\
64        && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
65      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
66        | xargs grep -H 'http:'\
67        | grep -v "| xargs grep -H 'htt"\
68        | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"
69      # Last grep just above will remove false positives from
70      # substitutions that fit on one line.
71
72      LFBFL_s_files_paths=$(
73        find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
74        | relevant_find\
75        | not_main_html_find
76      )
77      if [[ -z "${LFBFL_s_files_paths}" ]]; then
78        [[ LFBFL_i_verbose -eq 1 ]] && printf "No file found.\n"
79        continue
80      fi
81      mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
82      for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
83        [[ -f "${LFBFL_file_path}" ]] || continue
84        LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
85        [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
86          || continue
87        [[ LFBFL_i_verbose -eq 1 ]]\
88          && printf "Handling the file: %s.\n" "${LFBFL_file_path}"
89        for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
90          LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
91          if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
92            sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
93              "${LFBFL_file_path}"
94          fi
95        done
```

```
    done
  done
}
```

2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
```

```
# declare -r LFBFL_dependencies_URL="$2" too long
declare -r LFBFL_start_URL="$2"
local LFBFL_verbose=""
declare -i LFBFL_i_verbose=0
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
  LFBFL_i_verbose=1
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
declare -ar LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  # "$--work-directory={LFBFL_work_directory}" doesn't work
  "${LFBFL_work_directory_option}"
)

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
local LFBFL_file_name
local LFBFL_URL
local LFBFL_file_path

LFBFL_file_name="build_md_from_printable_md.exec.sh"
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='f94176cceb6fe386392c87e0c130611e4ac126323b6bc'
LFBFL_correct_sha512+='d5796ab82e9581f5749b7e4b0823f734085c8f6c22c7'
LFBFL_correct_sha512+='a7ed750718c040a3207a43d2321bf370ebc7287'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='883b5e6640475cda544fdb8191d444fc98a93c9c3545e'
LFBFL_correct_sha512+='5ad3769ba4187bea2addd3823a17e9d42bc1dd7056d1'
LFBFL_correct_sha512+='57517010a8094935211daad11fa7774a6651804'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

```

```
LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='99f80a0b9beb913094410e869454e952ea5b57f920673'
LFBFL_correct_sha512+='a58c3a3512e69300af8154e404ea5b935ac807c3e418'
LFBFL_correct_sha512+='d31b5dcfe77ad7335e8fbb5fa1511b5fa5d75d1'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='284e3bf4f66835020141ddb7d34fe5e36131c10b548cb'
LFBFL_correct_sha512+='5d34f59ae8fccbba4c827131eb7099323bc32a71c9b5'
LFBFL_correct_sha512+='7c0b9ef80fbccb385c3ce492d7a256f4743be42'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='b970e040536625ef0ac06bd10e6e164361c7fd89ae38f'
LFBFL_correct_sha512+='8a2a03033d221d1d6543e348cc98fb290e64ff26cdf7'
LFBFL_correct_sha512+='a8d225688a30bd0cbc16cf2a223ecb9e4954bfa'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='fd6b9dfb9da77447758f267ba51135f7279601852c90f'
LFBFL_correct_sha512+='630d25fb9408903ac3dbc9ecd0a5d25660d066650596'
LFBFL_correct_sha512+='445c3a10981d9df141ab2bb915bab5438690ccf'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='8b069a22b1e881fc3ccd9b339f1bedcbc3a5719870d4c'
LFBFL_correct_sha512+='56e33735d4514183afa2094db86859bfe29b05aad046'
```

```
  LFBFL_correct_sha512+='de81884cfc3268eaa818251ff92f5f15f42b686'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="generate_from_template.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='43b8d08be780b5cc9b0ba9fff339d23258be79e8ffc98'
  LFBFL_correct_sha512+='d983570e4245b946cb1540ceb3555e594d97d2dca213'
  LFBFL_correct_sha512+='d9a4d7a1d86c0bc3de455f6acb7489e19304d3f'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='fc041abf937df4c7cb1241c73e776e8bd07a5525bc9a9'
  LFBFL_correct_sha512+='29c83d597ddd6aa88a6e61a99d22ca88c11561f1cdbc'
  LFBFL_correct_sha512+='0fbf5df4794a2384f33e7e7dd5b65a53ab46f4a'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="grammar_and_spell_check.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='a34a57c303881c42c94db9d17fde07a6578839cc9e50b'
  LFBFL_correct_sha512+='07f26b1ef85f5bf19ccba74781fa83b7e69e0aab83a1'
  LFBFL_correct_sha512+='5d27e3231bce411f79da77dcccf4c6e482fcfe5'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  # /licenses_templates/ ---------------------------------------------
  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
  declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
  LFBFL_file_name="build_licenses_templates.exec.sh"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='68199cc1aa267c79691bb1c2339c60916efc0b50166c6'
  LFBFL_correct_sha512+='da9dc44f7fe3af7c238705e1c3aa75e2c4ed82e5bd5f'
  LFBFL_correct_sha512+='096f6190244583dbf589bcdc7a2623d05e56080'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="license_file_header_AGPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
  LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
  LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="license_file_header_GPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
  LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
  LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="license_file_header_LGPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
  LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
  LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  # ------------------------------------------------------------------

  LFBFL_file_name="lines_counts.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='eae635c1d61bf6d2bf9bf573a5f57e845f0261b2f3f16'
  LFBFL_correct_sha512+='81b7e27352f3fd3bf89eeec237ee4a0dfd24f1df2aa5'
  LFBFL_correct_sha512+='8953351b1c17d14988ade2d6a80fdd6099be98c'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="lines_filters.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='ec81a00c36795c8473157fb4c15eaaa36788b8d81517d'
LFBFL_correct_sha512+='338f97f2e959e796c8cc7cd66833f4a9728656037d15'
LFBFL_correct_sha512+='9a6a2b21e4692016b842c4df6d43861127f5979'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='f5641056e09a5f01ae9f4f432fd89283918f336b5b5b2'
LFBFL_correct_sha512+='f4514260ef11c142355acd7f879d4e12339344487e72'
LFBFL_correct_sha512+='c58eed79898f9a8a2ede83f4b49151f09da139a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='47029d60e49b85f5b96808358206176688ee0b62c6ac3'
LFBFL_correct_sha512+='5df6948ffcd53b8bdb7198170b7ce659ba44a4045d3d'
LFBFL_correct_sha512+='47dfee53cdd8149df76b12ba302fb2cfa6ec395'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='585809b2dd125cd45f4b3751fe7078f2b473fbec5af9c'
LFBFL_correct_sha512+='5d242b6dd827c680f22c913ddf9823c6d2225ffb0b29'
LFBFL_correct_sha512+='31865aa546aa5cc181f21224062bf04fc39c0f8'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="repository_data.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='793af27d8e658ed63148e6a3b45ba9b4eda1ddd3ede22'
  LFBFL_correct_sha512+='0b2af3c6a72e46d89b3565cef6e074cf1f9fcaccc7e4'
  LFBFL_correct_sha512+='dca98da471003bebeaabde704412ced55728094'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="shell_checks_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='37fad2bd06ef116694ed44721ea5d0d234d2268258916'
  LFBFL_correct_sha512+='3c4a285a096f116a8aa4098d1599eca7a3252778726c'
  LFBFL_correct_sha512+='e176132abf07dd1b861fcd338cc72630ef74934'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.exec.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
  LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
  LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.libr.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1413898a4fb4f31ca17993fd6110f7e1f32adf31c22c4'
  LFBFL_correct_sha512+='7fa46a31d3c1511a3112a611e04fac19bb838d4476d9'
  LFBFL_correct_sha512+='ecf77a28dd48d3d2732df7392c50e38262852d7'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="strings_functions.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
```

```
LFBFL_correct_sha512='21d587dba4893255af3dd7166324652ae80dee75a896b'
LFBFL_correct_sha512+='e2f685933a21a94e4f73fac80f428ec965e95e8e663a'
LFBFL_correct_sha512+='2596615c357f7f25047122fbe26d6a5362e08bf'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='9e00388c81114ec116fb4e0b8dd249480625b56b097fe'
LFBFL_correct_sha512+='8ae04e04d259443039cdabf7719ff12be99d07eee3a2'
LFBFL_correct_sha512+='e62e3b4dd6bcf08a2be2ffffe5093bed2fa711b'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='88e08eca01fdcbd28ed4abd6742c07f30d38352ffaaac'
LFBFL_correct_sha512+='26f65e78eab8b6ce734e648e9a7c0b64991a87eaa39f'
LFBFL_correct_sha512+='5c9eeeaf8a7eb663425f3f61f597eadfe81b6dc'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
```

```
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=shell_checks_complement.libr.sh
  source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

  enhanced_set_bash_option extglob
  # shellcheck source=repository_data.libr.sh
  source "./${LFBFL_subdir}/repository_data.libr.sh"
  # shellcheck disable=SC2154,SC2309
  [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
    && enhanced_unset_bash_option extglob

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
```

```
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        "--work-directory=${LFBFL_some_directory}"\
        "--base-name=${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_i_"

  declare -i LFBFL_i_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts-=LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
```

```
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options[@]}"

  printf -- "---Python---\n"
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
```

```
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
```

```
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
```

```
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file "${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"
```

```

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix="LFBFL_"
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options[@]}" \
    --max-line-length=${LFBFL_i_max_line_length}

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
```

```
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts-=LFBFL_i_vnu_jar_ts))
    ((LFBFL_i_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
```

```
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
```

```
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
  )

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  local LFBFL_file_name
  local LFBFL_URL
  local LFBFL_file_path

  LFBFL_file_name="build_md_from_printable_md.exec.sh"
  # LFBFL_script_download_URL
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_build_md_from_printable_md.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_check_shell_scripts_beginnings.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spell_check.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="repository_data.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_repository_data.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_shell_checks_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.exec.php@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.libr.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.libr.php@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_strings_functions.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_too_long_code_lines.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
261     chmod +x "./${LFBFL_file_path}"
262
263     # shellcheck source=check_shell_scripts_beginnings.libr.sh
264     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
265     # shellcheck source=check_shell_scripts_indentation.libr.sh
266     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
267     # shellcheck source=check_URLs.libr.sh
268     source "./${LFBFL_subdir}/check_URLs.libr.sh"
269     # shellcheck source=common_options.libr.sh
270     source "./${LFBFL_subdir}/common_options.libr.sh"
271     # shellcheck source=comparisons.libr.sh
272     source "./${LFBFL_subdir}/comparisons.libr.sh"
273     # shellcheck source=generate_from_template.libr.sh
274     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
275     # shellcheck source=get_common_text_glob_patterns.libr.sh
276     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
277     # shellcheck source=grammar_and_spell_check.libr.sh
278     source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
279     # shellcheck source=lines_counts.libr.sh
280     source "./${LFBFL_subdir}/lines_counts.libr.sh"
281     # shellcheck source=lines_filters.libr.sh
282     source "./${LFBFL_subdir}/lines_filters.libr.sh"
283     # shellcheck source=lines_maps.libr.sh
284     source "./${LFBFL_subdir}/lines_maps.libr.sh"
285     # shellcheck source=overwrite_if_not_equal.libr.sh
286     source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
287     # shellcheck source=python_black_complement.libr.sh
288     source "./${LFBFL_subdir}/python_black_complement.libr.sh"
289     # shellcheck source=python_isort_complement.libr.sh
290     source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
291     # shellcheck source=shell_checks_complement.libr.sh
292     source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
293     # shellcheck source=strings_functions.libr.sh
294     source "./${LFBFL_subdir}/strings_functions.libr.sh"
295     # shellcheck source=too_long_code_lines.libr.sh
296     source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"
297
298     enhanced_set_bash_option extglob
299     # shellcheck source=repository_data.libr.sh
300     source "./${LFBFL_subdir}/repository_data.libr.sh"
301     # shellcheck disable=SC2154,SC2309
302     [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
303       && enhanced_unset_bash_option extglob
304
305     enhanced_set_shell_option pipefail\
306       && trap 'enhanced_unset_shell_option pipefail' RETURN
```

```

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        "--work-directory=${LFBFL_some_directory}"\
        "--base-name=${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_i_"
```

```

  declare -i LFBFL_i_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts-=LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
```

```
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options[@]}"

  printf -- "---Python---\n"
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
```

```
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
```

```
LFBFL_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_phpmd_baseline

local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_temp_phpmd_baseline

local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
LFBFL_phpmd_rulesets+="design,naming,unusedcode"
readonly LFBFL_phpmd_rulesets

phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Saving new baseline in temp if necessary.
rm "${LFBFL_temp_phpmd_baseline}"
phpmd --color --generate-baseline\
  --baseline-file "${LFBFL_temp_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
printf -- "---PHP end---\n"

printf -- "---JS---\n"
LFBFL_npm_lint_directories=""
grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
  --result-variable-prefix="LFBFL_"
if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
  printf "Running ESLint\n"
  local LFBFL_JS_directory
  declare -a LFBFL_arr_paths
  mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
  for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
    (cd "${LFBFL_JS_directory}" && npm run lint)
  done
fi
printf -- "---JS end---\n"

printf "Analyzing too long lines\n"
# shellcheck disable=SC2248
too_long_code_lines "${LFBFL_some_common_options[@]}" \
  --max-line-length=${LFBFL_i_max_line_length}

printf "Analyzing shell scripts beginnings\n"
check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
  | relevant_grep

```

```
  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
```

```
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts-=LFBFL_i_vnu_jar_ts))
    ((LFBFL_i_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.8 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
```

```
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

```

```
  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

```

```
get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    readonly LFBFL_where_was_i
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --work-directory=* ]]; then
      LFBFL_work_directory=${LFBFL_arg#--work-directory=}
      LFBFL_work_directory=${LFBFL_work_directory:-.}
      # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
      LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
      break
    fi
  done
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
```

```
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
}

popd_from_work_directory(){
  enhanced_popd "${LFBFL_work_directory}" 3
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

```
enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
```

```
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

2.9 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
```

```
31    # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
32    # J'ai rerepensé au 3-way-merge
33    # et à mon article sur le principe de première différence
34    # appliqué à la largeur modulaire, de clique ou de rang avec
35    # l'idée d'un principe de première différence ternaire.
36    declare -i LFBFL_i_first=1
37    local LFBFL_current
38    local LFBFL_arg
39    for LFBFL_arg in "$@"; do
40      if [[ LFBFL_i_first -eq 1 ]]; then
41        LFBFL_current="${LFBFL_arg}"
42        LFBFL_i_first=0
43        continue
44      fi
45      if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
46        return 0
47      fi
48    done
49    return 1
50  }
51
52  some_distinct(){
53    ! equal "$@"
54  }
55
56  all_distinct(){
57    # Returns 1 if all the arguments strings are distinct.
58    declare -ir LFBFL_ij_max=$#
59    declare -i LFBFL_i
60    declare -i LFBFL_j
61    for ((LFBFL_i=1; LFBFL_i<=LFBFL_ij_max; ++LFBFL_i)); do
62      for ((LFBFL_j=1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
63        [[ LFBFL_i -eq LFBFL_j ]] && continue
64        [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
65      done
66    done
67    return 1
68  }
69
70  all_distinct2(){
71    # Returns 1 if all the arguments strings are distinct.
72    local LFBFL_element
73    local LFBFL_previous_element
74    if [[ $# -le 1 ]]; then
75      return 1
76    fi
```

```
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
```

```
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1"\
```

```
    | head -1
}

is_substring(){
  # $1=$string
  # $2=$substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=$file_name
  # $2=$sub_file_name
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

2.10 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
```

```
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1
```

```
53
54    enhanced_set_shell_option pipefail\
55      && trap 'enhanced_unset_shell_option pipefail' RETURN
56
57    local LFBFL_subdir2="build_and_checks_variables"
58    local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
59    local LFBFL_repository_name=""
60    grep_variable "${LFBFL_data_file_name}" repository_name\
61      --result-variable-prefix="LFBFL_"\
62      --replace-line-returns-by=""
63
64    local LFBFL_repository_web_url=""
65    grep_variable "${LFBFL_data_file_name}" repository_web_url\
66      --result-variable-prefix="LFBFL_"\
67      --replace-line-returns-by=""
68
69    local LFBFL_repository_git_url=""
70    grep_variable "${LFBFL_data_file_name}" repository_git_url\
71      --result-variable-prefix="LFBFL_"\
72      --replace-line-returns-by=""
73
74    local LFBFL_abstract=""
75    grep_variable "${LFBFL_data_file_name}" abstract\
76      --result-variable-prefix="LFBFL_"
77
78    local LFBFL_acknowledgments=""
79    grep_variable "${LFBFL_data_file_name}" acknowledgments\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_author_email=""
83    grep_variable "${LFBFL_data_file_name}" author_email\
84      --result-variable-prefix="LFBFL_"\
85      --replace-line-returns-by=""
86
87    local LFBFL_author_full_name=""
88    grep_variable "${LFBFL_data_file_name}" author_full_name\
89      --result-variable-prefix="LFBFL_"\
90      --replace-line-returns-by=" "
91
92    local LFBFL_author_website=""
93    grep_variable "${LFBFL_data_file_name}" author_website\
94      --result-variable-prefix="LFBFL_"\
95      --replace-line-returns-by=""
96
97    declare -i LFBFL_i_max_line_length
98    grep_variable "${LFBFL_data_file_name}" max_line_length\
```

```
    --result-variable-prefix="LFBFL_i_"

  declare -r LFBFL_current_date=$(date -I"date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_i_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc -l
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "$@"
    code_lines_count_not_empty "$@"
    code_lines_count_empty "$@"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  local LFBFL_temp_path
  # LFBFL_temp_path=$(realpath "./${LFBFL_subdir2}/temp") for pushd
  LFBFL_temp_path="./${LFBFL_subdir2}/temp"
  readonly LFBFL_temp_path

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree_light.txt"
```

```
145
146   tree -a -DFh --gitignore\
147     -I "node_modules/"\
148     -I "__pycache__/"\
149     -I ".mypy_cache/"\
150     -I ".ruff_cache/"\
151     -I ".git/"\
152     | replace_non_ascii_spaces\
153     > "${LFBFL_temp_path}/current_tree.txt.temp"
154
155   local LFBFL_s_files_paths
156   LFBFL_s_files_paths=$(
157     sed -Ez 's/(.)\\\n/\1/Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
158     | grep -v '^//'
159   )
160   readonly LFBFL_s_files_paths
161   declare -a LFBFL_arr_files_paths
162   if [[ -n "${LFBFL_s_files_paths}" ]]; then
163     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
164   fi
165   readonly LFBFL_arr_files_paths
166
167   local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
168   LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
169   readonly LFBFL_temp_files_listing
170   : > "${LFBFL_temp_files_listing}"
171   local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
172   LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
173   readonly LFBFL_temp_files_listing2
174   : > "${LFBFL_temp_files_listing2}"
175   # HTML <li> elements, hence "lis".
176   local LFBFL_temp_files_lis
177   LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.tpl"
178   readonly LFBFL_temp_files_lis
179   : > "${LFBFL_temp_files_lis}"
180   # shellcheck disable=SC2248
181   get_split_score_simple 1 ${LFBFL_i_max_line_length} /
182   declare -r LFBFL_score_command="${get_split_score_result}"
183   declare -i LFBFL_i_score_command_properties
184   LFBFL_i_score_command_properties="${i_get_split_score_result2}"
185   readonly LFBFL_i_score_command_properties
186   # shellcheck disable=SC2248
187   get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
188   declare -r LFBFL_score_command2="${get_split_score_result}"
189   declare -i LFBFL_i_score_command_properties2
190   LFBFL_i_score_command_properties2="${i_get_split_score_result2}"
```

```
191     readonly LFBFL_i_score_command_properties2
192     declare -r LFBFL_suffix='%'
193     declare -i LFBFL_i=0
194     local LFBFL_cleaned_path1
195     local LFBFL_cleaned_path2
196     local LFBFL_new_lines
197     local LFBFL_new_lines2
198     local LFBFL_new_lines3
199     local LFBFL_file_path
200     local LFBFL_s_format
201     # Remove line returns here to keep lines short.
202     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
203       printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
204       # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
205       # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
206       LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
207       LFBFL_cleaned_path2=$(
208         printf "%s" "${LFBFL_file_path}"\
209         | sed -e 's/\//:/g' -e 's/\.//g'
210       )
211       LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
212       if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
213         # shellcheck disable=SC2248
214         repeated_split_last_line "${LFBFL_new_lines}"\
215           ""\
216           ${LFBFL_i_max_line_length}\
217           "${LFBFL_suffix}"\
218           "${LFBFL_score_command}"\
219           "${LFBFL_i_score_command_properties}"\
220           "\\"
221         LFBFL_new_lines=${repeated_split_last_line_result}
222       fi
223       LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
224       if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
225         # shellcheck disable=SC2248
226         repeated_split_last_line "${LFBFL_new_lines2}"\
227           ""\
228           ${LFBFL_i_max_line_length}\
229           "${LFBFL_suffix}"\
230           "${LFBFL_score_command2}"\
231           "${LFBFL_i_score_command_properties2}"
232         LFBFL_new_lines2=${repeated_split_last_line_result}
233       fi
234       LFBFL_new_lines3="${LFBFL_file_path}"
235       if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
236         # shellcheck disable=SC2248
```

```
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_i_score_command_properties}"
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        "${LFBFL_i_score_command_properties}"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection2.%s\">2.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection2.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
```

```
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "${LFBFL_temp_path}/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_i_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_i_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(
      grep '.\{'${LFBFL_i_overlength}'\}' "${LFBFL_tree_path}"
    )
    if [[ -z "${LFBFL_s_lines}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)——[^—]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
```

```
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* (([a-zA-Z0-9\._/-]+).)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
```

```
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
```

```
# Tex filling:
if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
  sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
         -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
         -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
         -e "s|@author_email@|${LFBFL_author_email}|g"\
         -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
         -e "s|@author_website@|${LFBFL_author_website}|g"\
         -e "s|@current_date@|${LFBFL_current_date}|g"\
         -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
         -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
         -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
    "${LFBFL_tex_path_start}.1"

  insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
    "${LFBFL_temp_path}/abstract_temp"\
    "${LFBFL_tex_path_start}.2"

  insert_file_at_token "${LFBFL_tex_path_start}.2"\
    @acknowledgments@\
    "${LFBFL_temp_path}/acknowledgments_temp"\
    "${LFBFL_tex_path_start}.3"

  insert_file_at_token "${LFBFL_tex_path_start}.3"\
    @current_tree_light@\
    "${LFBFL_temp_path}/current_tree_light.txt"\
    "${LFBFL_tex_path_start}.4"

  insert_file_at_token "${LFBFL_tex_path_start}.4"\
    @current_tree@\
    "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_tex_path_start}.5"

  insert_file_at_token "${LFBFL_tex_path_start}.5"\
    @files_listing_VerbatimInput@\
    "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_tex_path_start}.6"

  overwrite_if_not_equal\
    "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
    "${LFBFL_tex_path_start}.6" 1
fi

# HTML filling:
declare -i LFBFL_i_HTML_updated=0
if [[ -f "${LFBFL_html_path_start}.1" ]]; then
```

```
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
           -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
           -e "s|@author_email@|${LFBFL_author_email}|g"\
           -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
           -e "s|@author_website@|${LFBFL_author_website}|g"\
           -e "s|@current_date@|${LFBFL_current_date}|g"\
           -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
           -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
           -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_html_path_start}.1"

    insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@ "${LFBFL_temp_files_lis}"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.tpl"\
      "${LFBFL_html_path_start}.7"

    overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.7" 1
    LFBFL_i_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
```

```
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
  fi

  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_name}" latex_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" './${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

2.11 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

```

```
generate_from_template_with_line_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    printf "%s\n" "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_part1
  # $4=$file_name_part2
  # Other arguments are options for grep.
  declare -a LFBFL_grep_options=()
  declare -i LFBFL_i=0
  local LFBFL_arg
  for ((LFBFL_i=5; LFBFL_i<=$#; ++LFBFL_i)); do
    LFBFL_arg="${!LFBFL_i}"
    LFBFL_grep_options+=("${LFBFL_arg}")
  done
  declare -ir LFBFL_i_line_number=$(
    grep -n "${LFBFL_grep_options[@]}" -- "$2" "$1"\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
  declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_i_lines_after=$((
    LFBFL_i_line_count - LFBFL_i_line_number
  ))
  head --lines="$((LFBFL_i_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_i_lines_after}" "$1"\
```

```
    > "$4"
}

insert_file_at_token(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_to_insert
  # $4=$file_name_result
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}" --fixed-strings
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}

```

```

get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
```

```
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spell_check(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_or_spell_checker_command=""
  grep_variable "$1" grammar_or_spell_checker_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
  declare -r LFBFL_command="${LFBFL_grammar_or_spell_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    return
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spell_check: No file found.\n"
      return
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  # Source
  local LFBFL_license_source_subdir
  LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
  readonly LFBFL_license_source_subdir
  local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
  LFBFL_source_license_prefix+="license_file_header_"
  readonly LFBFL_source_license_prefix
  # Target
  local LFBFL_license_target_subdir
```

```
62    LFBFL_license_target_subdir="${LFBFL_work_directory}/"
63    LFBFL_license_target_subdir="build_and_checks_variables/temp/"
64    readonly LFBFL_license_target_subdir
65    local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
66    LFBFL_target_license_prefix+="license_file_header_"
67    readonly LFBFL_target_license_prefix
68
69    declare -ar LFBFL_licenses=(
70      "AGPLv3+"
71      "GPLv3+"
72      "LGPLv3+"
73    )
74
75    # ------------------------------------------------------------------
76    # Various language definitions for text file formats
77    declare -ar LFBFL_block_comment_languages=(
78      "c"
79      "py"
80    )
81    declare -ar LFBFL_block_comment_enters=(
82      '/*'
83      '"""'
84    )
85    declare -ar LFBFL_block_comment_exits=(
86      '*/'
87      '"""'
88    )
89    equal "${#LFBFL_block_comment_languages[@]}"\
90          "${#LFBFL_block_comment_enters[@]}"\
91          "${#LFBFL_block_comment_exits[@]}"
92    declare -i LFBFL_i_equal_result=$?
93    if [[ LFBFL_i_equal_result -eq 0 ]]; then
94      printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
95      return 1
96    fi
97
98    declare -ar LFBFL_line_comment_languages=(
99      "sh"
100     "tex"
101   )
102   declare -ar LFBFL_line_comment_prefixes=(
103     '# '
104     '% '
105   )
106   equal "${#LFBFL_line_comment_languages[@]}"\
107         "${#LFBFL_line_comment_prefixes[@]}"
```

```
  LFBFL_i_equal_result=$?
  if [[ LFBFL_i_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
  local LFBFL_extension
  local LFBFL_license_file_name
  local LFBFL_enter_string
  local LFBFL_exit_string
  local LFBFL_temp2
  local LFBFL_prefix_string
  local LFBFL_intermediate_file_name
  local LFBFL_file_prefix
  declare -i LFBFL_i_generate_from_template_result
  for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
    LFBFL_license_prefix="${LFBFL_source_license_prefix}"
    LFBFL_license_prefix+="${LFBFL_some_license}"
    LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
    LFBFL_license_prefix2+="${LFBFL_some_license}"
    for ((i=0; i<${#LFBFL_block_comment_languages[@]}; i++)); do
      LFBFL_extension=${LFBFL_block_comment_languages[i]}
      LFBFL_license_file_name="${LFBFL_license_prefix2}"
      LFBFL_license_file_name+=".${LFBFL_extension}"
      LFBFL_enter_string=${LFBFL_block_comment_enters[i]}
      LFBFL_exit_string=${LFBFL_block_comment_exits[i]}
      generate_from_template_with_block_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_enter_string}" "${LFBFL_exit_string}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
```

```
154           printf "License template %s.tpl created.\n"\
155             "${LFBFL_license_file_name}"
156         fi
157       fi
158     done
159     LFBFL_temp2=".generate_from_template_with_line_comments.temp"
160     for ((i=0; i<${#LFBFL_line_comment_languages[@]}; i++)); do
161       LFBFL_extension=${LFBFL_line_comment_languages[i]}
162       LFBFL_license_file_name="${LFBFL_license_prefix2}"
163       LFBFL_license_file_name+=".${LFBFL_extension}"
164       LFBFL_prefix_string=${LFBFL_line_comment_prefixes[i]}
165       LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
166       LFBFL_intermediate_file_name+="${LFBFL_temp2}"
167       LFBFL_file_prefix=""
168       if [[ "${LFBFL_extension}" == "sh" ]]; then
169         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
170       fi
171       # The sed command is here to remove trailing spaces of previously
172       # empty lines after comment prefix is added to the line.
173       generate_from_template_with_line_comments\
174         "${LFBFL_license_prefix}.tpl"\
175         "${LFBFL_license_file_name}.tpl"\
176         "${LFBFL_prefix_string}"\
177         "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
178         "${LFBFL_file_prefix}"
179       LFBFL_i_generate_from_template_result=$?
180       if [[ LFBFL_i_verbose -eq 1 ]]; then
181         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
182           printf "License template %s.tpl updated.\n"\
183             "${LFBFL_license_file_name}"
184         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
185           printf "License template %s.tpl created.\n"\
186             "${LFBFL_license_file_name}"
187         fi
188       fi
189     done
190   done
191   # ------------------------------------------------------------------
192
193   # Now that base license templates generated languages license
194   # templates in working directory, we can go there.
195   pushd_to_work_directory\
196     && trap 'popd_from_work_directory' RETURN
197   can_continue_after_enhanced_pushd || return 1
198
199   local LFBFL_data_file_name="build_and_checks_variables/"
```

```
200   LFBFL_data_file_name+="repository_data.txt"
201   readonly LFBFL_data_file_name
202   local LFBFL_repository_name=""
203   grep_variable "${LFBFL_data_file_name}" repository_name\
204     --result-variable-prefix="LFBFL_"\
205     --replace-line-returns-by=""
206   local LFBFL_license=""
207   grep_variable "${LFBFL_data_file_name}" license\
208     --result-variable-prefix="LFBFL_"
209   local LFBFL_license2=""
210   grep_variable "${LFBFL_data_file_name}" license2\
211     --result-variable-prefix="LFBFL_"
212   local LFBFL_author_full_name=""
213   grep_variable "${LFBFL_data_file_name}" author_full_name\
214     --result-variable-prefix="LFBFL_"\
215     --replace-line-returns-by=" "
216
217   LFBFL_license_prefix="build_and_checks_variables/temp/"
218   LFBFL_license_prefix+="license_file_header_"
219   readonly LFBFL_license_prefix
220   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
221   readonly LFBFL_license_prefix2
222   declare -r\
223     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
224
225   # First year according to current state of git repository.
226   declare -ir LFBFL_i_first_year=$(
227     git log\
228     | grep 'Date:'\
229     | cut -f 8 -d ' '\
230     | tail -1
231   )
232   # Last year according to current state of git repository.
233   declare -ir LFBFL_i_last_year=$(
234     git log\
235     | grep 'Date:'\
236     | cut -f 8 -d ' '\
237     | head -1
238   )
239   local LFBFL_copyright_string
240   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
241   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
242   readonly LFBFL_copyright_string
243
244   prepare_filled_license_file_block(){
245     # $1=license_file_name
```

```
    sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
        -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
        "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf "License template %s updated.\n" "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
        printf "License template %s created.\n" "$1"
      fi
    fi
    head --lines=-1 "$1"\
      | tail --lines=+2\
      > "$1.temp"
    # Strange world: tail --lines=-1 was working
    # to exclude first line.
    # I tested it months ago, and saw with my eyes the start and end
    # block comments removed with:
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_name2
  declare -i LFBFL_i_not_subfile
  declare -i LFBFL_i_not_subfile2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
```

```
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}"\
        "${LFBFL_license_file_name}.temp"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}"\
          "${LFBFL_license_file_name2}.temp"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm "${LFBFL_license_file_name}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm "${LFBFL_license_file_name2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_name
    sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
        -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
        "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -ir LFBFL_i_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_i_result -eq 1 ]]; then
        printf "License template %s updated." "$1"
      elif [[ LFBFL_i_result -eq 2 ]]; then
```

```
338         printf "License template %s created." "$1"
339       fi
340     fi
341   }
342
343   declare -Ar LFBFL_all_line_comment_languages=(
344     ["sh"]="sh"
345     ["tex"]="tex"
346   )
347
348   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
349     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
350     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
351     prepare_filled_license_file_line "${LFBFL_license_file_name}"
352     if [[ -n "${LFBFL_license2}" ]]; then
353       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
354       LFBFL_license_file_name2+=".${LFBFL_dest}"
355       prepare_filled_license_file_line "${LFBFL_license_file_name2}"
356     fi
357
358     LFBFL_s_files_paths=$(
359       find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
360       | relevant_find
361     )
362     if [[ -z "${LFBFL_s_files_paths}" ]]; then
363       continue
364     fi
365     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
366     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
367       is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
368       LFBFL_i_not_subfile=$?
369       LFBFL_i_not_subfile2=1
370       if [[ -n "${LFBFL_license2}" ]]; then
371         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
372         LFBFL_i_not_subfile2=$?
373       fi
374       if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
375       then
376         printf "File %s has no/wrong license header?\n"\
377           "${LFBFL_file_path}"
378       fi
379     done
380   done
381 }
382
383 build_licenses_templates "$@"
```

2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
This file is part of @repository_name@ library.

@repository_name@ is free software:
you can redistribute it and/or modify it under the terms
of the GNU Affero General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

@repository_name@ is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Affero General Public License for more details.

You should have received a copy of
the GNU Affero General Public License
along with @repository_name@.
If not, see <https://www.gnu.org/licenses/>.

©Copyright @copyright_string@
```

2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

```
This file is part of @repository_name@ library.

@repository_name@ is free software:
you can redistribute it and/or modify it under the terms
of the GNU General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

@repository_name@ is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU General Public License for more details.

You should have received a copy of
the GNU General Public License
along with @repository_name@.
```

```
If not, see <https://www.gnu.org/licenses/>.

©Copyright @copyright_string@
```

2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

```
This file is part of @repository_name@ library.

@repository_name@ is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

@repository_name@ is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with @repository_name@.
If not, see <https://www.gnu.org/licenses/>.

©Copyright @copyright_string@
```

2.18 build_and_checks_dependencies/lines_counts.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H -v 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
```

```
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  # below and another lines, see SC2145 false positive
  all_code_lines "$@" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
```

```
  all_self_code_lines "$@" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi i_code_lines_count_all_result
  i_code_lines_count_all_result=$(
    all_self_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_all_result}
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi i_code_lines_count_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_empty_result}
```

```
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

2.19 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
```

```
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
```

```
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed -e 's|domain|lists.gnu.org|' \
#       -e 's|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
```

```
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent "${LFBFL_temp}" "${!#}"; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=filepath
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
```

```
    if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]];
    then
      LFBFL_i_quiet=1
      LFBFL_grep_options+=("${LFBFL_arg}")
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_arg}")
  done

  if [[
    LFBFL_i_enforce_line_starts -eq 0
    && LFBFL_i_enforce_line_ends -eq 0
    ]];
  then
    grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
    return
  fi
  declare -r LFBFL_s_lines=$(
    grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
      "$2" "$1"
  )
  declare -i LFBFL_i_result=1
  if [[ -z "${LFBFL_s_lines}" ]]; then
    # shellcheck disable=SC2248
    return ${LFBFL_i_result}
  fi
  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  declare -i LFBFL_i_fixed_string_length=${#2}
  local LFBFL_some_line
  local LFBFL_line_part
  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "$2" ]]; then
      [[ LFBFL_i_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
```

```
      LFBFL_i_result=0
      continue
    fi
    if [[
      LFBFL_i_enforce_line_starts -eq 1
      && LFBFL_i_enforce_line_ends -eq 1
      ]];
    then
      continue
    fi
    if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
    fi
    if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
      LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
    fi
    if [[ "${LFBFL_line_part}" == "$2" ]]; then
      [[ LFBFL_i_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_result=0
    fi
  done
  LFBFL_i_result=$((LFBFL_i_result | $?))
  # shellcheck disable=SC2248
  return ${LFBFL_i_result}
}

grep_variable(){
  # $1=$file
  # $2=$variable_name
  # Options :
  #   --replace-line-returns-by=""
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  local LFBFL_arg
  # printf "%s\n" "${LFBFL_regexp}"
  local LFBFL_variable_value
  if [[ "$*" == *--replace-line-returns-by=* ]]; then
    local LFBFL_s_replace=""
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
        LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
        break
      fi
    done
    LFBFL_variable_value=$(
      grep -oP "${LFBFL_regexp}" "$1"\
```

```
      | sed -Ez -e 's/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(grep -oP "${LFBFL_regexp}" "$1")
  fi
  local LFBFL_prefix=""
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep -v '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep -v '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep -E '^( |$)'
```

```
297 }
298
299 not_spampty_lines(){
300   grep '^[^ ]'
301   # grep -v -E '^( |$)'
302 }
303
304 all_space_lines(){
305   grep '^[ ]\+$'
306 }
307
308 not_all_space_lines(){
309   grep -v '^[ ]\+$'
310 }
311
312 # All space or empty lines = aspampty
313 aspampty_lines(){
314   grep '^[ ]*$'
315 }
316
317 not_aspampty_lines(){
318   grep '[^ ]'
319   # grep -v '^[ ]*$'
320 }
321
322 empty_lines_after_file_name(){
323   grep '^[^:]\+:$'
324 }
325
326 not_empty_lines_after_file_name(){
327   grep -v '^[^:]\+:$'
328 }
329
330 space_starting_lines_after_file_name(){
331   grep '^[^:]\+: '
332 }
333
334 not_space_starting_lines_after_file_name(){
335   grep -v '^[^:]\+: '
336 }
337
338 spampty_lines_after_file_name(){
339   grep -E '^[^:]+:( |$)'
340 }
341
342 not_spampty_lines_after_file_name(){
```

```
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep -v -E '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# :). Nice command :).

not_JS_dependencies_find(){
  grep -vE -e "(^|/)node_modules/"\
           -e "(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep -vE -e "^([^:]+/)?node_modules/"\
           -e "^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
```

```
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep -vE -e "(^|/)__pycache__/"\
           -e "(^|/)\.mypy_cache/"\
           -e "(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep -vE -e "^([^:]+/)?__pycache__/"\
           -e "^([^:]+/)?\.mypy_cache/"\
           -e "^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
}

not_git_find(){
  grep -vE "(^|/)\.git/"
}

not_git_grep(){
  grep -vE "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep -vE "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep -vE "^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep -vE -e "(^|/)COPYING$"\
           -e "(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep -vE -e "^([^:]+/)?COPYING:"\
```

```
            -e "^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep -vE "(^|/)build_and_checks_variables/temp/.*$"
}

not_temp_file_grep(){
  grep -vE "^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}
```

```

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

2.20 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
```

```
# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
```

```
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=$target_file_name
  # $2=$temp_file_name
  # $3=$keep_temp
  # $4=$tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories.
```

```
  # $5=--quiet : we even don't want the message of diff saying that
  #              files are distinct
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_i_is_equal
  if [[ -n "$4" ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff "$1" "$2"\
      | grep -E "^(>|<)"\
      | grep -v "/$"
    LFBFL_i_is_equal=${PIPESTATUS[2]}
  else
    if [[ -n "$5" ]]; then
      diff -q "$1" "$2" > /dev/null
    else
      diff -q "$1" "$2"
    fi
    LFBFL_i_is_equal=1-$?
  fi
  if [[ LFBFL_i_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

2.22 build_and_checks_dependencies/python_black_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
```

```
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  check_no_empty_line_after_python_function_docstring "$@"
}
```

2.23 build_and_checks_dependencies/python_isort_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking import of _collections_abc is at the right place\n"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.py" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_collections_abc_place: No file found.\n"
```

```
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ -f "${LFBFL_file_path}" ]] || continue
    sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      "${LFBFL_file_path}"\
      > "${LFBFL_file_path}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_path}"\
      "${LFBFL_file_path}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  check_collections_abc_place "$@"
}
```

2.24 build_and_checks_dependencies/repository_data.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
```

```
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_name
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_bash_option extglob\
    && trap 'enhanced_unset_bash_option extglob' RETURN

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
```

```
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

2.25 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1
```

```
41
42   enhanced_set_bash_option globstar\
43     && trap 'enhanced_unset_bash_option globstar' RETURN
44
45   local LFBFL_s="Checking that shell scripts do not"
46   LFBFL_s+=" contain wrong array size expansion."
47   [[ LFBFL_i_verbose -eq 1 ]]\
48     && printf "%s\n" "${LFBFL_s}"
49
50   pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
51 }
52
53 check_no_misplaced_then(){
54   # Options:
55   #   --verbose
56   #   --work-directory=""
57   #   --max-line-length
58   declare -i LFBFL_i_verbose=0
59   get_verbose_option "$@"
60   local LFBFL_work_directory=""
61   get_work_directory_option "$@"
62   pushd_to_work_directory\
63     && trap 'popd_from_work_directory' RETURN
64   can_continue_after_enhanced_pushd || return 1
65
66   local LFBFL_arg
67   declare -i LFBFL_i_max_line_length=70
68   for LFBFL_arg in "$@"; do
69     if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
70       LFBFL_arg=${LFBFL_arg#--max-line-length=}
71       LFBFL_i_max_line_length=$((LFBFL_arg))
72       break
73     fi
74   done
75   declare -i LFBFL_i_max_previous_line_length
76   LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))
77
78   enhanced_set_bash_option globstar\
79     && trap 'enhanced_unset_bash_option globstar' RETURN
80
81   [[ LFBFL_i_verbose -eq 1 ]]\
82     && printf\
83         "Checking that shell scripts do not contain misplaced 'then'.\n"
84
85   local LFBFL_regexp
86   LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
```

```
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"

  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  check_no_size_of_array_first_element "$@"
  check_no_misplaced_then "$@"
}
```

2.26 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
```

```
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

2.27 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
```

```
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  $b_after_before,
  $i_max_length,
  $a_delimiter_strings_domain,
){
  return function(
    $s_delimiter_string,
    $i_cut_position,
    $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain){
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
```

```
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}
```

2.28 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
```

```
36
37 split_line_at(){
38   # $1=$line
39   # $2=$position : number of characters in split_line_at_result_beginning
40   declare -g split_line_at_result_beginning="${1:0:$2}"
41   declare -g split_line_at_result_end="${1:$2}"
42   # printf "%s\n" "$split_line_at_result_beginning"
43   # printf "%s\n" "$split_line_at_result_end"
44 }
45
46 #-----------------------------------------------------------------------
47 #-----------------------------------------------------------------------
48 # Properties for the score for a possible split/cut in the string between
49 # two characters.
50 #-----------------------------------------------------------------------
51 # Some preliminary remarks:
52 # a) If the character before the split position is a forbidden previous
53 #   character, the score is -1.
54 #   (Usually, either we have no or one forbidden previous character that is
55 #   a backslash.)
56 # b) If the previous character is not forbidden,
57 #   and either the character before the position is among the delimiters,
58 #   or the character after the position is among the delimiters,
59 #   then the score is computed according to the delimiter found,
60 #   the position, and the fact that the position is after or before the
61 #   delimiter.
62 # c) If two consecutive delimiters are found in the string,
63 #   the maximum score between the one after the first delimiter
64 #   and before the second delimiter is taken for the score between the
65 #   two delimiters.
66 # d) Otherwise, by default,
67 #   the score for cutting/splitting at some position is 0.
68 # e) A substring is given by the goal length given for the resulting
69 #   strings/lines after split.
70 # f) When some scores are equal,
71 #   we split at the rightmost position with maximum score in the substring.
72 # g) Remark f) implies that below we do not need to consider strict
73 #    inequalities for the score.
74 # h) The score function SF(position) which will dictate our choice for
75 #   split/cut position is in reality
76 #   SF1a(
77 #     position, previous_character, next_character, delimiters, forbiddens
78 #   )
79 #   but at the cost of expliciting its internal and adding verbosity,
80 #   we can see that the score really depends on the following arguments
81 #   SF1b(
```

```
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
```

```
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
```

```
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 + 1 and RP7 : p2 = p1 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logic relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that's not an additional
#   property that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
```

```
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
```

```
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     for all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
```

```
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
```

```
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
```

```
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
```

```
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
```

```
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
    | SSP_LARGER_AFTER
    | SSP_LARGER_BEFORE
    | SSP_POSITION_NOT_DECREASING
    | SSP_POSITION_NOT_DECREASING_AFTER
    | SSP_POSITION_NOT_DECREASING_BEFORE
    | SSP_POSITION_UNIFORM
    | SSP_POSITION_UNIFORM_AFTER
    | SSP_POSITION_UNIFORM_BEFORE
    | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
    | SSP_SAME_DELIMITER_LARGER_AFTER
    | SSP_SAME_DELIMITER_LARGER_BEFORE
    | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
    | SSP_SAME_POSITION_LARGER_AFTER
    | SSP_SAME_POSITION_LARGER_BEFORE
```

```
542       | SSP_UNIFORM
543     ))
544     until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
545       LFBFL_i_result_old=$((LFBFL_i_result))
546       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
547         LFBFL_i_result=$((
548           LFBFL_i_result
549           | SSP_LARGER_AFTER
550           | SSP_SAME_DELIMITER_LARGER_AFTER
551           | SSP_SAME_POSITION_LARGER_AFTER
552         ))
553       fi
554       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
555         LFBFL_i_result=$((
556           LFBFL_i_result
557           | SSP_LARGER_BEFORE
558           | SSP_SAME_DELIMITER_LARGER_BEFORE
559           | SSP_SAME_POSITION_LARGER_BEFORE
560         ))
561       fi
562       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
563         LFBFL_i_result=$((
564           LFBFL_i_result
565           | SSP_POSITION_NOT_DECREASING_AFTER
566           | SSP_POSITION_NOT_DECREASING_BEFORE
567         ))
568       fi
569       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
570         LFBFL_i_result=$((
571           LFBFL_i_result
572           | SSP_POSITION_UNIFORM_AFTER
573           | SSP_POSITION_UNIFORM_BEFORE
574           | SSP_POSITION_NOT_DECREASING
575           | SSP_POSITION_NOT_DECREASING_AFTER
576           | SSP_POSITION_NOT_DECREASING_BEFORE
577         ))
578       fi
579       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
580         LFBFL_i_result=$((
581           LFBFL_i_result
582           | SSP_POSITION_NOT_DECREASING_AFTER
583         ))
584       fi
585       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
586         LFBFL_i_result=$((
587           LFBFL_i_result
```

```
588         | SSP_POSITION_NOT_DECREASING_BEFORE
589       ))
590     fi
591     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
592       LFBFL_i_result=$((
593         LFBFL_i_result
594         | SSP_EQUAL_AFTER_OR_BEFORE
595         | SSP_LARGER_AFTER
596         | SSP_LARGER_BEFORE
597         | SSP_SAME_DELIMITER_LARGER_AFTER
598         | SSP_SAME_DELIMITER_LARGER_BEFORE
599       ))
600     fi
601     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
602       LFBFL_i_result=$((
603         LFBFL_i_result
604         | SSP_LARGER_AFTER
605       ))
606     fi
607     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
608       LFBFL_i_result=$((
609         LFBFL_i_result
610         | SSP_LARGER_BEFORE
611       ))
612     fi
613     if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
614       LFBFL_i_result=$((
615         LFBFL_i_result
616         | SSP_EQUAL_AFTER_OR_BEFORE
617         | SSP_LARGER_AFTER
618         | SSP_LARGER_BEFORE
619         | SSP_SAME_POSITION_LARGER_AFTER
620         | SSP_SAME_POSITION_LARGER_BEFORE
621       ))
622     fi
623     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
624       LFBFL_i_result=$((
625         LFBFL_i_result
626         | SSP_LARGER_AFTER
627       ))
628     fi
629     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
630       LFBFL_i_result=$((
631         LFBFL_i_result
632         | SSP_LARGER_BEFORE
633       ))
```

```
634       fi
635       if ((LFBFL_i_result & SSP_UNIFORM)); then
636         LFBFL_i_result=$((LFBFL_i_all_flags))
637         break
638       fi
639       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
640       && ((
641         LFBFL_i_result & (
642           SSP_ALWAYS_LARGER_BEFORE
643           | SSP_EQUAL_AFTER_OR_BEFORE
644           | SSP_LARGER_BEFORE
645           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
646           | SSP_SAME_DELIMITER_LARGER_BEFORE
647           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
648           | SSP_SAME_POSITION_LARGER_BEFORE
649         )
650       )); then
651         LFBFL_i_result=$((LFBFL_i_all_flags))
652         break
653       fi
654       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
655       && ((
656         LFBFL_i_result & (
657           SSP_EQUAL_AFTER_OR_BEFORE
658           | SSP_LARGER_AFTER
659           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
660           | SSP_SAME_DELIMITER_LARGER_AFTER
661           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
662           | SSP_SAME_POSITION_LARGER_AFTER
663         )
664       )); then
665         LFBFL_i_result=$((LFBFL_i_all_flags))
666         break
667       fi
668       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
669       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
670         LFBFL_i_result=$((
671           LFBFL_i_result
672           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
673         ))
674       fi
675       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
676       && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
677         LFBFL_i_result=$((
678           LFBFL_i_result
679           | SSP_SAME_POSITION_LARGER_AFTER
```

```
680         ))
681       fi
682       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
683       && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
684         LFBFL_i_result=$((
685           LFBFL_i_result
686           | SSP_SAME_POSITION_LARGER_BEFORE
687         ))
688       fi
689       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
690       && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
691         LFBFL_i_result=$((LFBFL_i_all_flags))
692         break
693       fi
694       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
695       && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
696         LFBFL_i_result=$((
697           LFBFL_i_result
698           | SSP_ALWAYS_LARGER_AFTER
699         ))
700       fi
701       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
702       && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
703         LFBFL_i_result=$((
704           LFBFL_i_result
705           | SSP_ALWAYS_LARGER_BEFORE
706         ))
707       fi
708       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
709       && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
710         LFBFL_i_result=$((
711           LFBFL_i_result
712           | SSP_EQUAL_AFTER_OR_BEFORE
713         ))
714       fi
715       if ((LFBFL_i_result & SSP_LARGER_AFTER))\
716       && ((
717         LFBFL_i_result & (
718           SSP_POSITION_UNIFORM
719           | SSP_POSITION_UNIFORM_AFTER
720         )
721       )); then
722         LFBFL_i_result=$((
723           LFBFL_i_result
724           | SSP_SAME_DELIMITER_LARGER_AFTER
725         ))
```

```
726     fi
727     if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
728     && ((
729       LFBFL_i_result & (
730         SSP_POSITION_UNIFORM
731         | SSP_POSITION_UNIFORM_BEFORE
732       )
733     )); then
734       LFBFL_i_result=$((
735         LFBFL_i_result
736         | SSP_SAME_DELIMITER_LARGER_BEFORE
737       ))
738     fi
739     if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
740     && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
741       LFBFL_i_result=$((
742         LFBFL_i_result
743         | SSP_POSITION_NOT_DECREASING
744       ))
745     fi
746     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
747     && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
748       LFBFL_i_result=$((
749         LFBFL_i_result
750         | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
751       ))
752     fi
753     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
754     && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
755       LFBFL_i_result=$((LFBFL_i_all_flags))
756       break
757     fi
758     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
759     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
760       LFBFL_i_result=$((
761         LFBFL_i_result
762         | SSP_ALWAYS_LARGER_AFTER
763       ))
764     fi
765     if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
766     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
767       LFBFL_i_result=$((
768         LFBFL_i_result
769         | SSP_ALWAYS_LARGER_BEFORE
770       ))
771     fi
```

```
772     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
773     && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
774       LFBFL_i_result=$((
775         LFBFL_i_result
776         | SSP_POSITION_UNIFORM
777       ))
778     fi
779     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
780     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
781       LFBFL_i_result=$((
782         LFBFL_i_result
783         | SSP_ALWAYS_LARGER_AFTER
784       ))
785     fi
786     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
787     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
788       LFBFL_i_result=$((
789         LFBFL_i_result
790         | SSP_ALWAYS_LARGER_BEFORE
791       ))
792     fi
793     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
794     && ((
795       LFBFL_i_result & (
796         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
797         | SSP_SAME_POSITION_LARGER_AFTER
798         | SSP_SAME_POSITION_LARGER_BEFORE
799       )
800     )); then
801       LFBFL_i_result=$((LFBFL_i_all_flags))
802       break
803     fi
804     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
805     && ((
806       LFBFL_i_result & (
807         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
808         | SSP_SAME_POSITION_LARGER_BEFORE
809       )
810     )); then
811       LFBFL_i_result=$((LFBFL_i_all_flags))
812       break
813     fi
814     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
815     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
816       LFBFL_i_result=$((
817         LFBFL_i_result
```

```
818         | SSP_ALWAYS_LARGER_AFTER
819       ))
820     fi
821     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
822     && ((
823       LFBFL_i_result & (
824         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
825         |  SSP_SAME_POSITION_LARGER_AFTER
826       )
827     )); then
828       LFBFL_i_result=$((LFBFL_i_all_flags))
829       break
830     fi
831     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
832     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
833       LFBFL_i_result=$((
834         LFBFL_i_result
835         | SSP_ALWAYS_LARGER_BEFORE
836       ))
837     fi
838   done
839   declare -gi i_split_score_properties_logical_closure_result=$((
840     LFBFL_i_result
841   ))
842 }
843
844 split_score_simple(){
845   # $1=$after_before
846   # $2=$delimiters_strings_domain concatenated characters/delimiters
847   # $3=$delimiter_string a single character
848   # $4=$is_cut_after
849   declare -gi i_split_score_result
850   if [[ "$2" != *$3* ]]; then
851     i_split_score_result=0
852     return
853   fi
854   if [[ "$4" == "$1" ]]; then
855     i_split_score_result=2
856     return
857   fi
858   i_split_score_result=1
859 }
860
861 get_split_score_simple(){
862   # $1=$after_before
863   # $2=$max_length
```

```
  # $3=$delimiters_strings_domain
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((11-$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   i_get_split_score_result2=7
  # else
  #   i_get_split_score_result2=11
  # fi
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_POSITION_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
  ))
}

get_split_score_exec(){
  declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
  SPLIT_SCORE_EXEC+="/split_score.exec.php"
}

split_score(){
  # $1=$after_before
  # $2=$max_length
  # $3=$delimiters_strings_domain concatenated characters/delimiters
  # $4=$delimiter_string a single character
  # $5=$cut_position
  # $6=$is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi i_split_score_result
  i_split_score_result=$(eval "${LFBFL_command}")
}

# This function is currently not called,
```

```
# hence we need that shellcheck disable.
# shellcheck disable=SC2034
get_split_score(){
  # $1=$after_before
  # $1=$max_length
  # $2=$delimiters_strings_domain
  declare -g get_split_score_result="split_score $1 $2 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((9-$1*4))
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
  ))
}

split_line_at_most(){
  # $1=$line
  # $2=$max_length of beginning result string
  # $3=$split_score_command
  #   negative score means do not consider as a possible split
  # $4=$split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=$forbidden_previous_character
  # $6=$split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
```

```
#     in that case, only mono-character strings are considered,
#   - this argument can be an array of characters,
#     optimize accordingly,
#   - this argument can be an array of strings,
#     optimize accordingly,
#   - this argument can be an array of regexps,
#     optimize accordingly.
#     Not sure an array of regexps makes more sense than a single
#     one. Maybe they could be both cases with corresponding
#     optimizations. Same for a single character or substring.
#
# Note that we're always looking for the last position to split
# among maximum score;
# if some position gets as score from 2 delimiters,
# its score is always the maximum of the two corresponding scores.
declare -g split_line_at_most_result_start
declare -g split_line_at_most_result_end
declare -A LFBFL_positions
# printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
# For my use case in bash scripts, I will need only an array of
# characters. See get_split_score().
if [[ -n "$6" ]]; then
  printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
fi
if [[ $2 -ge ${#1} ]]; then
  split_line_at_most_result_start=$1
  split_line_at_most_result_end=""
  return
fi
# We know that ${#1} > $2, hence ${#1} - 1 >= $2
# Edge cases:
# There is always at least one character that is overlength,
# but we need to look only at the score before that character that is
# overlength, the score after that character is invalid,
# whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
# That first character that is overlength has index $2.
# The cut position after that character has index $2 + 1.
# Looping from character position going from 0 to $2, we need to drop
# score before character 0 and drop score after character $2.
# We do this with forbidden score -1.
declare -ir LFBFL_i_max=$(($2))
declare -i LFBFL_i
declare -i LFBFL_j
local LFBFL_current_char

# At the beginning, each position has score 0.
```

```
1002    LFBFL_positions["0"]="-1"
1003    for ((LFBFL_i=1; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
1004      LFBFL_positions["${LFBFL_i}"]="0"
1005    done
1006    LFBFL_j=$((LFBFL_i+1))
1007    LFBFL_positions["${LFBFL_j}"]="-1"
1008
1009    # If we have a forbidden previous character,
1010    # we mark the forbidden positions with -1.
1011    if [[ -n "$5" ]]; then
1012      for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
1013        LFBFL_j=$((LFBFL_i+1))
1014        LFBFL_current_char="${1:${LFBFL_i}:1}"
1015        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1016          LFBFL_positions["${LFBFL_j}"]="-1"
1017        fi
1018      done
1019    fi
1020
1021    declare -r LFBFL_sort_command="sort --numeric-sort"
1022
1023    if (($4 & SSP_DELIMITER_UNIFORM))\
1024    && (($4 & SSP_LARGER_AFTER))\
1025    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1026      for ((LFBFL_i=LFBFL_i_max; LFBFL_i>=0; --LFBFL_i)) do
1027        LFBFL_j=$((LFBFL_i+1))
1028        LFBFL_current_char="${1:${LFBFL_i}:1}"
1029        eval "$3 '${LFBFL_current_char}' 1"
1030        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1031        if [[ i_split_score_result -ge 1 ]]; then
1032          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1033          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1034            LFBFL_positions["${LFBFL_j}"]=$(
1035              max "${LFBFL_sort_command}"\
1036                "${LFBFL_positions["${LFBFL_j}"]}"\
1037                "${i_split_score_result}"
1038            )
1039            break
1040          fi
1041        fi
1042        eval "$3 '${LFBFL_current_char}' 0"
1043        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1044        if [[ i_split_score_result -ge 1 ]]; then
1045          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1046          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1047            LFBFL_positions["${LFBFL_i}"]=$(
```

```
1048            max "${LFBFL_sort_command}"\
1049              "${LFBFL_positions["${LFBFL_i}"]}"\
1050              "${i_split_score_result}"
1051          )
1052        fi
1053      fi
1054    done
1055  elif (($4 & SSP_DELIMITER_UNIFORM))\
1056  && (($4 & SSP_LARGER_BEFORE))\
1057  && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1058    for ((LFBFL_i=LFBFL_i_max; LFBFL_i>=0; --LFBFL_i)) do
1059      LFBFL_j=$((LFBFL_i+1))
1060      LFBFL_current_char="${1:${LFBFL_i}:1}"
1061      # We still need to break in second if, after computing score after,
1062      # because SSP_LARGER_BEFORE implies large inequality instead of
1063      # strict inequality.
1064      # And since the algorithm is to take the largest position with
1065      # maximum score, there is a possibility that we'll break on a found
1066      # score before but for splitting on a score after.
1067      eval "$3 '${LFBFL_current_char}' 1"
1068      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1069      if [[ i_split_score_result -ge 1 ]]; then
1070        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1071        if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1072          LFBFL_positions["${LFBFL_j}"]=$(
1073            max "${LFBFL_sort_command}"\
1074              "${LFBFL_positions["${LFBFL_j}"]}"\
1075              "${i_split_score_result}"
1076          )
1077        fi
1078      fi
1079      eval "$3 '${LFBFL_current_char}' 0"
1080      # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1081      if [[ i_split_score_result -ge 1 ]]; then
1082        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1083        if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1084          LFBFL_positions["${LFBFL_i}"]=$(
1085            max "${LFBFL_sort_command}"\
1086              "${LFBFL_positions["${LFBFL_i}"]}"\
1087              "${i_split_score_result}"
1088          )
1089          break
1090        fi
1091      fi
1092    done
1093  else
```

```
1094      for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
1095        LFBFL_j=$((LFBFL_i+1))
1096        LFBFL_current_char="${1:${LFBFL_i}:1}"
1097        eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1098        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1099        if [[ i_split_score_result -ge 1 ]]; then
1100          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1101          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1102            LFBFL_positions["${LFBFL_i}"]=$(
1103              max "${LFBFL_sort_command}"\
1104                "${LFBFL_positions["${LFBFL_i}"]}"\
1105                "${i_split_score_result}"
1106            )
1107          fi
1108        fi
1109        eval "$3 '${LFBFL_current_char}' ${LFBFL_j} 1"
1110        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1111        if [[ i_split_score_result -ge 1 ]]; then
1112          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1113          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1114            LFBFL_positions["${LFBFL_j}"]=$(
1115              max "${LFBFL_sort_command}"\
1116                "${LFBFL_positions["${LFBFL_j}"]}"\
1117                "${i_split_score_result}"
1118            )
1119          fi
1120        fi
1121      done
1122    fi
1123    declare -ir LFBFL_i_max_score=$(
1124      max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1125    )
1126    local LFBFL_value
1127    declare -i LFBFL_i_best_position
1128    for ((LFBFL_i=LFBFL_i_max; LFBFL_i>=0; --LFBFL_i)) do
1129      if [[ LFBFL_i -eq 0 ]]; then
1130        printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
1131      fi
1132      local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
1133      # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
1134      if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
1135        LFBFL_i_best_position=$((LFBFL_i))
1136        break
1137      fi
1138    done
1139    # shellcheck disable=SC2248
```

```
  split_line_at "$1" ${LFBFL_i_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=$new_lines
  # $2=$prefix
  # $3=$max_length
  # $4=$suffix
  # $5=$split_score_command
  # $6=$split_score_command_properties
  # $7=$forbidden_previous_character ('\' usually)
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_i_max_length_plus=$(($3 + 1))
  declare -ir LFBFL_i_length2=$(($3 - ${#4}))
  declare -r LFBFL_regexp='.\{'"${LFBFL_i_max_length_plus}"'\}$'

  # Testing if some line is too long.
  printf "%s" "$1"\
```

```
    | grep -q "${LFBFL_regexp}";
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    printf "%s" "$1"\
    | head --lines=-1
  )
  # printf "start: %s\n" "${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    printf "%s" "$1"\
    | tail --lines=1
  )
  # printf "last_line: %s\n" "${LFBFL_last_line}"
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    split_line_at_most "${LFBFL_last_line}" "${LFBFL_i_length2}"\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
```

```
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

2.29 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "too_long_code_lines.sh"
```

```
25 # to "too_long_code_lines.libr.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.libr.sh
29 source "./${LFBFL_subdir}/common_options.libr.sh"
30 # shellcheck source=get_common_text_glob_patterns.libr.sh
31 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
32 # shellcheck source=lines_filters.libr.sh
33 source "./${LFBFL_subdir}/lines_filters.libr.sh"
34
35 too_long_code_lines(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   #   --max-line-length
40   declare -i LFBFL_i_verbose=0
41   get_verbose_option "$@"
42   local LFBFL_work_directory=""
43   get_work_directory_option "$@"
44   pushd_to_work_directory\
45     && trap 'popd_from_work_directory' RETURN
46   can_continue_after_enhanced_pushd || return 1
47
48   local LFBFL_arg
49   declare -i LFBFL_i_max_line_length=70
50   for LFBFL_arg in "$@"; do
51     if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
52       LFBFL_arg=${LFBFL_arg#--max-line-length=}
53       LFBFL_i_max_line_length=$((LFBFL_arg))
54       break
55     fi
56   done
57
58   enhanced_set_shell_option pipefail\
59     && trap 'enhanced_unset_shell_option pipefail' RETURN
60
61   get_COMMON_TEXT_FILES_GLOB_PATTERNS
62   local LFBFL_pattern
63   local LFBFL_long_line
64   local LFBFL_file_name
65   local LFBFL_line
66   local LFBFL_extension
67   local LFBFL_base_name
68   declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
69   local LFBFL_s_long_lines
70   declare -a LFBFL_arr_long_lines
```

```
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | xargs grep -H '.\{'${LFBFL_i_overlength}'\}'
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_name=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_name##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_name%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.tpl"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

2.30 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp "./${LFBFL_common_file_name}.tpl"\
```

```
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_file_names=(
    "build_md_from_printable_md.exec.sh"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spell_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
    "overwrite_if_not_equal.libr.sh"
    "python_black_complement.libr.sh"
    "python_isort_complement.libr.sh"
    "repository_data.libr.sh"
    "shell_checks_complement.libr.sh"
    "split_score.exec.php"
    "split_score.libr.php"
    "strings_functions.libr.sh"
    "too_long_code_lines.libr.sh"
  )

  local LFBFL_file_name
  local LFBFL_file_sha512
  local LFBFL_file_sha512_all
  local LFBFL_base_file_name
  for LFBFL_file_name in "${LFBFL_file_names[@]}"; do
    LFBFL_file_sha512=$(
      sha512sum "./${LFBFL_file_name}"\
      | cut -f1 -d' '
    )
    LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
```

```
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

2.31 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_write=0
  if [[ "$*" == *--write* ]]; then
```

```
56      LFBFL_i_write=1
57    fi
58    readonly LFBFL_i_write
59
60    declare -i LFBFL_i_append=0
61    if [[ "$*" == *--append* ]]; then
62      LFBFL_i_append=1
63    fi
64    readonly LFBFL_i_append
65
66    enhanced_set_shell_option pipefail\
67      && trap 'enhanced_unset_shell_option pipefail' RETURN
68
69    LFBFL_subdir2="build_and_checks_variables"
70    local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
71    readonly LFBFL_listing
72    local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
73    readonly LFBFL_data_file_name
74
75    declare -i LFBFL_i_max_line_length
76    grep_variable "${LFBFL_data_file_name}" max_line_length\
77      --result-variable-prefix="LFBFL_i_"
78
79    if [[ LFBFL_i_write -eq 1 ]]; then
80      : > "${LFBFL_listing}"
81    fi
82    # Remove line returns that are here to keep lines short.
83    sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
84      > "${LFBFL_listing}.temp"
85    # shopt -s dotglob was needed at some point but I don't see why now.
86    # shellcheck disable=SC2248
87    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
88    declare -r LFBFL_split_score_command="${get_split_score_result}"
89    declare -i LFBFL_i_split_score_command_properties
90    LFBFL_i_split_score_command_properties="${i_get_split_score_result2}"
91    readonly LFBFL_i_split_score_command_properties
92    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
93    readonly LFBFL_suffix
94    # There is always a $'\n' final suffix from >>.
95    # But when origin string ends with the suffix we must add something.
96
97    local LFBFL_s_file_paths
98    LFBFL_s_file_paths=$(
99      find . -type f -printf '%P\n'\
100     | relevant_find\
101     | sort
```

```
102   )
103   declare -a LFBFL_arr_file_paths
104   if [[ -n "${LFBFL_s_files_paths}" ]]; then
105     mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
106   fi
107
108   declare -r LFBFL_final_suffix=$'\n\\'
109   local LFBFL_file_path
110   local LFBFL_base_file_name
111   for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
112     git check-ignore -q "${LFBFL_file_path}" && continue
113     LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
114     [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
115       || continue
116     if [[ "${LFBFL_base_file_name}" == *.md ]]; then
117       if [[ -f "${LFBFL_file_path}.tpl" ]]; then
118         continue
119       fi
120     fi
121     if [[ LFBFL_i_verbose -eq 1 ]]; then
122       printf "Non-ignored file: %s\n" "${LFBFL_file_path}"
123     fi
124     # shellcheck disable=SC2248
125     repeated_split_last_line "${LFBFL_file_path}"\
126       ""\
127       ${LFBFL_i_max_line_length}\
128       "${LFBFL_suffix}"\
129       "${LFBFL_split_score_command}"\
130       "${LFBFL_i_split_score_command_properties}"\
131       ""\
132       "${LFBFL_final_suffix}"
133     if [[ LFBFL_i_write -eq 1 ]]; then
134       # shellcheck disable=SC2001
135       printf "%s\n" "${repeated_split_last_line_result}"\
136         >> "${LFBFL_listing}"
137       continue
138     fi
139     if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
140         "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
141     then
142       :
143     else
144       printf "The file %s is not listed in %s.\n"\
145         "${LFBFL_file_path}"\
146         "${LFBFL_listing}"
147       if [[ LFBFL_i_append -eq 1 ]]; then
```

```
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_file_paths=$(cat "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
  fi

  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    if ! [[ -f "${LFBFL_file_path}" ]]; then
      printf "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

2.32 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

2.33 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

2.34 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

2.35 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

2.36 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
```

```
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
```

```
86    content: counter(level1) "." counter(level2) " ";
87    counter-increment: level2;
88  }
89  </style>
90  <title>@repository_name@</title>
91  </head>
92  <body >
93  <table class="title">
94    <tr>
95      <td style="padding:1ex">
96        <h1 class="titlemain">@repository_name@</h1>
97        <h2 class="titlerest">
98          @author_full_name@
99          <br>
100         <a href="@author_website@">
101           <span style="font-family:monospace">
102           @author_website@
103           </span>
104         </a>
105         <br>
106         <a href="mailto:@author_email@">
107           <span style="font-family:monospace">
108           @author_email@
109           </span>
110         </a>
111       </h2>
112     </td>
113   </tr>
114 </table>
115
116 <blockquote class="abstract">
117   <span style="font-weight:bold">Abstract:</span>
118   @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
123 <p>
124 Code lines: @number_of_lines@
125 </p>
126
127
128 <h2>Contents</h2>
129
130 <ol class="level1">
131   <li><a href="#section1">Files Tree</a></li>
```

```
  <li><a href="#section2">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Listing of files</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
```

```
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

<p>
Bash can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/bash/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/bash/</span></a>,<br>
<a href="https://tiswww.case.edu/php/chet/bash/bashtop.html"><span
  style="font-family:monospace"
>https://tiswww.case.edu/php/chet/bash/bashtop.html</span></a>,<br>
<a href="https://savannah.gnu.org/projects/bash/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/bash/</span></a>.
</p>

<p>
GNU coreutils can be found at the following URLs:<br>
```

```
<a href="https://www.gnu.org/software/coreutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/coreutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/coreutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/coreutils/</span></a>.
</p>

<p>
uutils coreutils can be found at the following URLs:<br>
<a href="https://uutils.github.io/coreutils/"><span
  style="font-family:monospace"
>https://uutils.github.io/coreutils/</span></a>,<br>
<a href="https://github.com/uutils/coreutils"><span
  style="font-family:monospace"
>https://github.com/uutils/coreutils</span></a>.
</p>

<p>
diff (GNU diffutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/diffutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/diffutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/diffutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/diffutils/</span></a>.
</p>

<p>
find, xargs (GNU findutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/findutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/findutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/findutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/findutils/</span></a>.
</p>

<p>
grep (GNU grep) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/grep/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/grep/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/grep/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/grep/</span></a>.
```

```
</p>

<p>
sed (GNU sed) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/sed/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/sed/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/sed/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/sed/</span></a>.
</p>

<p>
ShellCheck can be found at the following URLs:<br>
<a href="https://www.shellcheck.net/"><span
  style="font-family:monospace"
>https://www.shellcheck.net/</span></a>,<br>
<a href="https://github.com/koalaman/shellcheck"><span
  style="font-family:monospace"
>https://github.com/koalaman/shellcheck</span></a>.
</p>

<p>
GNU Wget can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/wget/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/wget/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/wget/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/wget/</span></a>.
</p>

<p>
more of util-linux can be found at the following URLs:<br>
<a href="https://www.kernel.org/pub/linux/utils/util-linux/"><span
  style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span></a>,<br>
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
><span
  style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
></a>,<br>
<a href="https://github.com/util-linux/util-linux"><span
  style="font-family:monospace"
>https://github.com/util-linux/util-linux</span></a> official?
```

```
</p>

<p>
pandoc can be found at the following URLs:<br>
<a href="https://pandoc.org/"><span
  style="font-family:monospace"
>https://pandoc.org/</span></a>,<br>
<a href="https://github.com/jgm/pandoc"><span
  style="font-family:monospace"
>https://github.com/jgm/pandoc</span></a>.
</p>

<p>
pdftex/pdflatex can be found at the following URLs:<br>
<a href="https://www.tug.org/applications/pdftex/"><span
  style="font-family:monospace"
>https://www.tug.org/applications/pdftex/</span></a>,<br>
<a href="https://tug.org/texlive/svn/"><span
  style="font-family:monospace"
>https://tug.org/texlive/svn/</span></a>.
</p>

<p>
tree can be found at the following URLs:<br>
<a href="https://oldmanprogrammer.net/source.php?dir=projects/tree"
><span style="font-family:monospace"
>https://oldmanprogrammer.net/source.php?dir=projects/tree</span></a>,<br>
<a href="https://gitlab.com/OldManProgrammer/unix-tree"><span
  style="font-family:monospace"
>https://gitlab.com/OldManProgrammer/unix-tree</span></a>,<br>
<a href="https://github.com/Old-Man-Programmer/tree"><span
  style="font-family:monospace"
>https://github.com/Old-Man-Programmer/tree</span></a>.
</p>

<p>
git can be found at the following URLs:<br>
<a href="https://git-scm.com/about"><span
  style="font-family:monospace"
>https://git-scm.com/about</span></a>,<br>
<a href="https://github.com/git/git"><span
  style="font-family:monospace"
>https://github.com/git/git</span></a>.
</p>

<p>
```

```
pylint can be found at the following URLs:<br>
<a href="https://www.pylint.org/"><span
  style="font-family:monospace"
>https://www.pylint.org/</span></a>,<br>
<a href="https://pylint.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://pylint.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/pylint-dev/pylint"><span
  style="font-family:monospace"
>https://github.com/pylint-dev/pylint</span></a>,<br>
<a href="https://pypi.org/project/pylint/"><span
  style="font-family:monospace"
>https://pypi.org/project/pylint/</span></a>.
</p>

<p>
notify-send can be found at the following URLs:<br>
<a href="https://gitlab.gnome.org/GNOME/libnotify/"><span
  style="font-family:monospace"
>https://gitlab.gnome.org/GNOME/libnotify/</span></a>,<br>
<a href="https://gnome.pages.gitlab.gnome.org/libnotify/"><span
  style="font-family:monospace"
>https://gnome.pages.gitlab.gnome.org/libnotify/</span></a>.
</p>

<p>
black can be found at the following URLs:<br>
<a href="https://github.com/psf/black"><span
  style="font-family:monospace"
>https://github.com/psf/black</span></a>,<br>
<a href="https://black.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://black.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://pypi.org/project/black/"><span
  style="font-family:monospace"
>https://pypi.org/project/black/</span></a>.
</p>

<p>
isort can be found at the following URLs:<br>
<a href="https://pycqa.github.io/isort/"><span
  style="font-family:monospace"
>https://pycqa.github.io/isort/</span></a>,<br>
<a href="https://github.com/pycqa/isort/"><span
  style="font-family:monospace"
>https://github.com/pycqa/isort/</span></a>,<br>
```

```
<a href="https://pypi.org/project/isort/"><span
  style="font-family:monospace"
>https://pypi.org/project/isort/</span></a>.
</p>

<p>
mypy can be found at the following URLs:<br>
<a href="https://mypy-lang.org/"><span
  style="font-family:monospace"
>https://mypy-lang.org/</span></a>,<br>
<a href="https://mypy.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://mypy.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/python/mypy"><span
  style="font-family:monospace"
>https://github.com/python/mypy</span></a>,<br>
<a href="https://pypi.org/project/mypy/"><span
  style="font-family:monospace"
>https://pypi.org/project/mypy/</span></a>.
</p>

<p>
pip can be found at the following URLs:<br>
<a href="https://pip.pypa.io/en/stable/"><span
  style="font-family:monospace"
>https://pip.pypa.io/en/stable/</span></a>,<br>
<a href="https://github.com/pypa/pip"><span
  style="font-family:monospace"
>https://github.com/pypa/pip</span></a>,<br>
<a href="https://pypi.org/project/pip/"><span
  style="font-family:monospace"
>https://pypi.org/project/pip/</span></a> (egg and chicken ;) ).
</p>

<p>
ruff can be found at the following URLs:<br>
<a href="https://docs.astral.sh/ruff/"><span
  style="font-family:monospace"
>https://docs.astral.sh/ruff/</span></a>,<br>
<a href="https://github.com/astral-sh/ruff"><span
  style="font-family:monospace"
>https://github.com/astral-sh/ruff</span></a>,<br>
<a href="https://pypi.org/project/ruff/"><span
  style="font-family:monospace"
>https://pypi.org/project/ruff/</span></a>.
</p>
```

```
<p>
bandit can be found at the following URLs:<br>
<a href="https://bandit.readthedocs.io/en/latest/"><span
  style="font-family:monospace"
>https://bandit.readthedocs.io/en/latest/</span></a>,<br>
<a href="https://github.com/PyCQA/bandit"><span
  style="font-family:monospace"
>https://github.com/PyCQA/bandit</span></a>,<br>
<a href="https://pypi.org/project/bandit/"><span
  style="font-family:monospace"
>https://pypi.org/project/bandit/</span></a>.
</p>

<p>
phpmd can be found at the following URLs:<br>
<a href="https://phpmd.org/"><span
  style="font-family:monospace"
>https://phpmd.org/</span></a>,<br>
<a href="https://github.com/phpmd/phpmd"><span
  style="font-family:monospace"
>https://github.com/phpmd/phpmd</span></a>,<br>
<a href="https://packagist.org/packages/phpmd/phpmd"><span
  style="font-family:monospace"
>https://packagist.org/packages/phpmd/phpmd</span></a>.
</p>

<p>
php can be found at the following URLs:<br>
<a href="https://www.php.net/"><span
  style="font-family:monospace"
>https://www.php.net/</span></a>,<br>
<a href="https://github.com/php/php-src"><span
  style="font-family:monospace"
>https://github.com/php/php-src</span></a>.
</p>

<p>
tex can be found at the following URL:<br>
<a href="https://ctan.org/tex-archive/systems/knuth/dist/tex"><span
  style="font-family:monospace"
>https://ctan.org/tex-archive/systems/knuth/dist/tex</span></a>.
</p>

<p>
pcregrep, pcre2grep can be found at the following URLs:<br>
```

```
<a href="https://www.pcre.org/"><span
  style="font-family:monospace"
>https://www.pcre.org/</span></a>,<br>
<a href="https://pcre2project.github.io/pcre2/"><span
  style="font-family:monospace"
>https://pcre2project.github.io/pcre2/</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre1"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre1</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre2"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2</span></a>,<br>
<a
href="https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c</span
></a>.
</p>

<p>
python3 can be found at the following URLs:<br>
<a href="https://www.python.org/"><span
  style="font-family:monospace"
>https://www.python.org/</span></a>,<br>
<a href="https://docs.python.org/3/"><span
  style="font-family:monospace"
>https://docs.python.org/3/</span></a>.
</p>

<p>
If you see something that I missed regarding my dependencies,
please tell me/email me :).
</p>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
```

```
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

2.37 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

```

```
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@
```

```
Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Listing of files}
\label{section:listing}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
```

```
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:\newline
\url{https://www.gnu.org/software/bash/},\newline
\url{https://tiswww.case.edu/php/chet/bash/bashtop.html},\newline
\url{https://savannah.gnu.org/projects/bash/}.

GNU coreutils can be found at the following URLs:\newline
\url{https://www.gnu.org/software/coreutils/},\newline
\url{https://savannah.gnu.org/projects/coreutils/}.

uutils coreutils can be found at the following URLs:\newline
\url{https://uutils.github.io/coreutils/},\newline
\url{https://github.com/uutils/coreutils}.

diff (GNU diffutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/diffutils/},\newline
\url{https://savannah.gnu.org/projects/diffutils/}.

find, xargs (GNU findutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/findutils/},\newline
\url{https://savannah.gnu.org/projects/findutils/}.

grep (GNU grep) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/grep/},\newline
\url{https://savannah.gnu.org/projects/grep/}.

```

```
sed (GNU sed) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/sed/},\newline
\url{https://savannah.gnu.org/projects/sed/}.

ShellCheck can be found at the following URLs:\newline
\url{https://www.shellcheck.net/},\newline
\url{https://github.com/koalaman/shellcheck}.

GNU Wget can be found at the following URLs:\newline
\url{https://www.gnu.org/software/wget/},\newline
\url{https://savannah.gnu.org/projects/wget/}.

more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},
\newline
\url{https://github.com/util-linux/util-linux} official?

pandoc can be found at the following URLs:\newline
\url{https://pandoc.org/},\newline
\url{https://github.com/jgm/pandoc}.

pdftex/pdflatex can be found at the following URLs:\newline
\url{https://www.tug.org/applications/pdftex/},\newline
\url{https://tug.org/texlive/svn/}.

tree can be found at the following URLs:\newline
\url{https://oldmanprogrammer.net/source.php?dir=projects/tree},\newline
\url{https://gitlab.com/OldManProgrammer/unix-tree},\newline
\url{https://github.com/Old-Man-Programmer/tree}.

git can be found at the following URLs:\newline
\url{https://git-scm.com/about},\newline
\url{https://github.com/git/git}.

pylint can be found at the following URLs:\newline
\url{https://www.pylint.org/},\newline
\url{https://pylint.readthedocs.io/en/stable/},\newline
\url{https://github.com/pylint-dev/pylint},\newline
\url{https://pypi.org/project/pylint/}.

notify-send can be found at the following URLs:\newline
\url{https://gitlab.gnome.org/GNOME/libnotify/},\newline
\url{https://gnome.pages.gitlab.gnome.org/libnotify/}.
```

```
black can be found at the following URLs:\newline
\url{https://github.com/psf/black},\newline
\url{https://black.readthedocs.io/en/stable/},\newline
\url{https://pypi.org/project/black/}.

isort can be found at the following URLs:\newline
\url{https://pycqa.github.io/isort/},\newline
\url{https://github.com/pycqa/isort/},\newline
\url{https://pypi.org/project/isort/}.

mypy can be found at the following URLs:\newline
\url{https://mypy-lang.org/},\newline
\url{https://mypy.readthedocs.io/en/stable/},\newline
\url{https://github.com/python/mypy},\newline
\url{https://pypi.org/project/mypy/}.

pip can be found at the following URLs:\newline
\url{https://pip.pypa.io/en/stable/},\newline
\url{https://github.com/pypa/pip},\newline
\url{https://pypi.org/project/pip/} (egg and chicken ;) ).

ruff can be found at the following URLs:\newline
\url{https://docs.astral.sh/ruff/},\newline
\url{https://github.com/astral-sh/ruff},\newline
\url{https://pypi.org/project/ruff/}.

bandit can be found at the following URLs:\newline
\url{https://bandit.readthedocs.io/en/latest/},\newline
\url{https://github.com/PyCQA/bandit},\newline
\url{https://pypi.org/project/bandit/}.

phpmd can be found at the following URLs:\newline
\url{https://phpmd.org/},\newline
\url{https://github.com/phpmd/phpmd},\newline
\url{https://packagist.org/packages/phpmd/phpmd}.

php can be found at the following URLs:\newline
\url{https://www.php.net/},\newline
\url{https://github.com/php/php-src}.

tex can be found at the following URL:\newline
\url{https://ctan.org/tex-archive/systems/knuth/dist/tex}.

pcregrep, pcre2grep can be found at the following URLs:\newline
\url{https://www.pcre.org/},\newline
```

```
\url{https://pcre2project.github.io/pcre2/},\newline
\url{https://github.com/PCRE2Project/pcre1},\newline
\url{https://github.com/PCRE2Project/pcre2},\newline
{\small
  \url{https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c}
}.

python3 can be found at the following URLs:\newline
\url{https://www.python.org/},\newline
\url{https://docs.python.org/3/}.

If you see something that I missed regarding my dependencies,
please tell me/email me :).

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

2.38 build_and_checks_variables/files_listing.tex.tpl

```
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


```

```
\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}
```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/grammar\_and\_spell\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spell_checklibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
```

```
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
```

```
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}

```

```latex

\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/repository\_data.libr.sh
}
\label{
  build_and_checks_dependencies:repository_datalibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/repository_data.libr.sh
```

```
}


\subsection{
  build\_and\_checks\_dependencies/shell\_checks\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:shell_checks_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/shell_checks_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}


\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
\label{
  build_and_checks_dependencies:strings_functionslibrsh
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}


\subsection{
  build\_and\_checks\_variables/.gitignore
}
```

```
\label{
  build_and_checks_variables:gitignore
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}


\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
}


\subsection{
```

```
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_names\_listing.txt
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
}

```

```latex

\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}


\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
```

```
}


\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}


\subsection{
  build\_and\_checks.exec.sh
}
\label{
  build_and_checksexecsh
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}


\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/git_checks.libr.sh
}


\subsection{
  git\_helpers/post-commit
}
\label{
  git_helpers:post-commit
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}


\subsection{
  LiensUtiles.txt
}
```

```
\label{
  LiensUtilestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  miscellaneous/funny.sh
}
\label{
  miscellaneous:funnysh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/funny.sh
}


\subsection{
  miscellaneous/spampty\_data.txt
}
\label{
  miscellaneous:spampty_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/spampty_data.txt
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
}


\subsection{
```

```
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}


\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}


```

2.39 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
```

```
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
```

```
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
```

```
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901234\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345678901123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

2.40 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
```

```
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.


Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/
```

```

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
```

```
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
```

```
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

2.41 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"/>
</phpmd-baseline>
```

2.42 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
```

```
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

2.43 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
```

```
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_or_spell_checker_command=more


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
```

```
              "3.3.3" : {}
            }
          }
        },
        "D": {
          "0.1.0": {
            "C": {
              // Dumb, you did not found it by yourself? Failed exam.
              "4.4.4" : {}
            }
          }
        }
      }
    },
    "B": {
      "2.2.2": {
        "C": {
          // Dumb, you did not found it by yourself? Failed exam.
          "4.4.4" : {}
        }
      }
    }
  }
  Wonderful someone rediscovered nested structures XD
  Seriously so many engineers or computer scientists are dumb to cry.

  bandit_venv=../venv_bandit
  black_venv=../venv_black
  isort_venv=../venv_isort
  mypy_venv=../venv_mypy
  pylint_venv=../venv_pylint
  ruff_venv=../venv_ruff


  upgrade venvs can be auto or ask, any other value is interpreted
  as ask.

  upgrade_venvs=ask


  time interval can be 0 to always upgrade or ask to upgrade,
  or a number at most bash max int
  https://superuser.com/questions/1030122/
  what-is-the-maximum-value-of-a-numeric-bash-shell-variable
  it should be enough to simulate never "upgrade or ask for upgrade"
  unless you claim to live many billions years ;)
```

```
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800
```

2.44 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

```

```
  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

2.45 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

2.46 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
```

```
            False positive when c contains \n.
- SC2034 : Unused variable
            False positive for globals that are declared in functions
            that are not currently called.
            Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
            Use printf '..%s..' "$foo".
            False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
            False positive when returning integer variable
            via an indirection like return ${!foo}.
            See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
            False positive when passing arguments would yield
            a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
            False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
            False positive when assigning a variable declared with -i.
            https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
            False positive when using "$@" line return and pipes.
            https://github.com/koalaman/shellcheck/issues/3192
            Example:
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
            Very useful but false positive when I play with
            variable variable names.
- SC2162X: read without -r will mangle backslashes.
            False positive when read is used without variable:
            https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
            False positive on returning integer values.
            https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
              Use = to compare as string (or expand explicitly with $var).
            Very useful but false positive when I play with
            variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
            value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
            https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
```

```
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

2.47 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
```

```
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\n\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1008c3559f993b4d8d9cbd6bfc1e21d02ab662d249632'
  LFBFL_correct_sha512+='5a3c4d7f1de0e3da96f9ced493a051b3c97ba02a1d49'
  LFBFL_correct_sha512+='332777524b0bdcfbfce0f8695aee572ca3d9614'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
```

```
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.48 COPYING

```
                    GNU GENERAL PUBLIC LICENSE
                       Version 3, 29 June 2007

 Copyright (C) 2007 Free Software Foundation, Inc. <https://fsf.org/>
 Everyone is permitted to copy and distribute verbatim copies
 of this license document, but changing it is not allowed.

                            Preamble

  The GNU General Public License is a free, copyleft license for
software and other kinds of works.

  The licenses for most software and other practical works are designed
to take away your freedom to share and change the works.  By contrast,
the GNU General Public License is intended to guarantee your freedom to
share and change all versions of a program--to make sure it remains free
software for all its users.  We, the Free Software Foundation, use the
```

GNU General Public License for most of our software; it applies also to any other work released this way by its authors. You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for them if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs, and that you know you can do these things.

To protect your rights, we need to prevent others from denying you these rights or asking you to surrender the rights. Therefore, you have certain responsibilities if you distribute copies of the software, or if you modify it: responsibilities to respect the freedom of others.

For example, if you distribute copies of such a program, whether gratis or for a fee, you must pass on to the recipients the same freedoms that you received. You must make sure that they, too, receive or can get the source code. And you must show them these terms so they know their rights.

Developers that use the GNU GPL protect your rights with two steps: (1) assert copyright on the software, and (2) offer you this License giving you legal permission to copy, distribute and/or modify it.

For the developers' and authors' protection, the GPL clearly explains that there is no warranty for this free software. For both users' and authors' sake, the GPL requires that modified versions be marked as changed, so that their problems will not be attributed erroneously to authors of previous versions.

Some devices are designed to deny users access to install or run modified versions of the software inside them, although the manufacturer can do so. This is fundamentally incompatible with the aim of protecting users' freedom to change the software. The systematic pattern of such abuse occurs in the area of products for individuals to use, which is precisely where it is most unacceptable. Therefore, we have designed this version of the GPL to prohibit the practice for those products. If such problems arise substantially in other domains, we stand ready to extend this provision to those domains in future versions of the GPL, as needed to protect the freedom of users.

Finally, every program is threatened constantly by software patents. States should not allow patents to restrict development and use of software on general-purpose computers, but in those that do, we wish to

avoid the special danger that patents applied to a free program could make it effectively proprietary. To prevent this, the GPL assures that patents cannot be used to render the program non-free.

The precise terms and conditions for copying, distribution and modification follow.

TERMS AND CONDITIONS

0. Definitions.

"This License" refers to version 3 of the GNU General Public License.

"Copyright" also means copyright-like laws that apply to other kinds of works, such as semiconductor masks.

"The Program" refers to any copyrightable work licensed under this License. Each licensee is addressed as "you". "Licensees" and "recipients" may be individuals or organizations.

To "modify" a work means to copy from or adapt all or part of the work in a fashion requiring copyright permission, other than the making of an exact copy. The resulting work is called a "modified version" of the earlier work or a work "based on" the earlier work.

A "covered work" means either the unmodified Program or a work based on the Program.

To "propagate" a work means to do anything with it that, without permission, would make you directly or secondarily liable for infringement under applicable copyright law, except executing it on a computer or modifying a private copy. Propagation includes copying, distribution (with or without modification), making available to the public, and in some countries other activities as well.

To "convey" a work means any kind of propagation that enables other parties to make or receive copies. Mere interaction with a user through a computer network, with no transfer of a copy, is not conveying.

An interactive user interface displays "Appropriate Legal Notices" to the extent that it includes a convenient and prominently visible feature that (1) displays an appropriate copyright notice, and (2) tells the user that there is no warranty for the work (except to the extent that warranties are provided), that licensees may convey the work under this License, and how to view a copy of this License. If the interface presents a list of user commands or options, such as a

menu, a prominent item in the list meets this criterion.

  1. Source Code.

  The "source code" for a work means the preferred form of the work for making modifications to it. "Object code" means any non-source form of a work.

  A "Standard Interface" means an interface that either is an official standard defined by a recognized standards body, or, in the case of interfaces specified for a particular programming language, one that is widely used among developers working in that language.

  The "System Libraries" of an executable work include anything, other than the work as a whole, that (a) is included in the normal form of packaging a Major Component, but which is not part of that Major Component, and (b) serves only to enable use of the work with that Major Component, or to implement a Standard Interface for which an implementation is available to the public in source code form. A "Major Component", in this context, means a major essential component (kernel, window system, and so on) of the specific operating system (if any) on which the executable work runs, or a compiler used to produce the work, or an object code interpreter used to run it.

  The "Corresponding Source" for a work in object code form means all the source code needed to generate, install, and (for an executable work) run the object code and to modify the work, including scripts to control those activities. However, it does not include the work's System Libraries, or general-purpose tools or generally available free programs which are used unmodified in performing those activities but which are not part of the work. For example, Corresponding Source includes interface definition files associated with source files for the work, and the source code for shared libraries and dynamically linked subprograms that the work is specifically designed to require, such as by intimate data communication or control flow between those subprograms and other parts of the work.

  The Corresponding Source need not include anything that users can regenerate automatically from other parts of the Corresponding Source.

  The Corresponding Source for a work in source code form is that same work.

  2. Basic Permissions.

All rights granted under this License are granted for the term of copyright on the Program, and are irrevocable provided the stated conditions are met. This License explicitly affirms your unlimited permission to run the unmodified Program. The output from running a covered work is covered by this License only if the output, given its content, constitutes a covered work. This License acknowledges your rights of fair use or other equivalent, as provided by copyright law.

You may make, run and propagate covered works that you do not convey, without conditions so long as your license otherwise remains in force. You may convey covered works to others for the sole purpose of having them make modifications exclusively for you, or provide you with facilities for running those works, provided that you comply with the terms of this License in conveying all material for which you do not control copyright. Those thus making or running the covered works for you must do so exclusively on your behalf, under your direction and control, on terms that prohibit them from making any copies of your copyrighted material outside their relationship with you.

Conveying under any other circumstances is permitted solely under the conditions stated below. Sublicensing is not allowed; section 10 makes it unnecessary.

3. Protecting Users' Legal Rights From Anti-Circumvention Law.

No covered work shall be deemed part of an effective technological measure under any applicable law fulfilling obligations under article 11 of the WIPO copyright treaty adopted on 20 December 1996, or similar laws prohibiting or restricting circumvention of such measures.

When you convey a covered work, you waive any legal power to forbid circumvention of technological measures to the extent such circumvention is effected by exercising rights under this License with respect to the covered work, and you disclaim any intention to limit operation or modification of the work as a means of enforcing, against the work's users, your or third parties' legal rights to forbid circumvention of technological measures.

4. Conveying Verbatim Copies.

You may convey verbatim copies of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice; keep intact all notices stating that this License and any non-permissive terms added in accord with section 7 apply to the code;

keep intact all notices of the absence of any warranty; and give all recipients a copy of this License along with the Program.

You may charge any price or no price for each copy that you convey, and you may offer support or warranty protection for a fee.

5. Conveying Modified Source Versions.

You may convey a work based on the Program, or the modifications to produce it from the Program, in the form of source code under the terms of section 4, provided that you also meet all of these conditions:

a) The work must carry prominent notices stating that you modified it, and giving a relevant date.

b) The work must carry prominent notices stating that it is released under this License and any conditions added under section 7. This requirement modifies the requirement in section 4 to "keep intact all notices".

c) You must license the entire work, as a whole, under this License to anyone who comes into possession of a copy. This License will therefore apply, along with any applicable section 7 additional terms, to the whole of the work, and all its parts, regardless of how they are packaged. This License gives no permission to license the work in any other way, but it does not invalidate such permission if you have separately received it.

d) If the work has interactive user interfaces, each must display Appropriate Legal Notices; however, if the Program has interactive interfaces that do not display Appropriate Legal Notices, your work need not make them do so.

A compilation of a covered work with other separate and independent works, which are not by their nature extensions of the covered work, and which are not combined with it such as to form a larger program, in or on a volume of a storage or distribution medium, is called an "aggregate" if the compilation and its resulting copyright are not used to limit the access or legal rights of the compilation’s users beyond what the individual works permit. Inclusion of a covered work in an aggregate does not cause this License to apply to the other parts of the aggregate.

6. Conveying Non-Source Forms.

You may convey a covered work in object code form under the terms

of sections 4 and 5, provided that you also convey the machine-readable Corresponding Source under the terms of this License, in one of these ways:

a) Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by the Corresponding Source fixed on a durable physical medium customarily used for software interchange.

b) Convey the object code in, or embodied in, a physical product (including a physical distribution medium), accompanied by a written offer, valid for at least three years and valid for as long as you offer spare parts or customer support for that product model, to give anyone who possesses the object code either (1) a copy of the Corresponding Source for all the software in the product that is covered by this License, on a durable physical medium customarily used for software interchange, for a price no more than your reasonable cost of physically performing this conveying of source, or (2) access to copy the Corresponding Source from a network server at no charge.

c) Convey individual copies of the object code with a copy of the written offer to provide the Corresponding Source. This alternative is allowed only occasionally and noncommercially, and only if you received the object code with such an offer, in accord with subsection 6b.

d) Convey the object code by offering access from a designated place (gratis or for a charge), and offer equivalent access to the Corresponding Source in the same way through the same place at no further charge. You need not require recipients to copy the Corresponding Source along with the object code. If the place to copy the object code is a network server, the Corresponding Source may be on a different server (operated by you or a third party) that supports equivalent copying facilities, provided you maintain clear directions next to the object code saying where to find the Corresponding Source. Regardless of what server hosts the Corresponding Source, you remain obligated to ensure that it is available for as long as needed to satisfy these requirements.

e) Convey the object code using peer-to-peer transmission, provided you inform other peers where the object code and Corresponding Source of the work are being offered to the general public at no charge under subsection 6d.

A separable portion of the object code, whose source code is excluded

from the Corresponding Source as a System Library, need not be included in conveying the object code work.

A "User Product" is either (1) a "consumer product", which means any tangible personal property which is normally used for personal, family, or household purposes, or (2) anything designed or sold for incorporation into a dwelling. In determining whether a product is a consumer product, doubtful cases shall be resolved in favor of coverage. For a particular product received by a particular user, "normally used" refers to a typical or common use of that class of product, regardless of the status of the particular user or of the way in which the particular user actually uses, or expects or is expected to use, the product. A product is a consumer product regardless of whether the product has substantial commercial, industrial or non-consumer uses, unless such uses represent the only significant mode of use of the product.

"Installation Information" for a User Product means any methods, procedures, authorization keys, or other information required to install and execute modified versions of a covered work in that User Product from a modified version of its Corresponding Source. The information must suffice to ensure that the continued functioning of the modified object code is in no case prevented or interfered with solely because modification has been made.

If you convey an object code work under this section in, or with, or specifically for use in, a User Product, and the conveying occurs as part of a transaction in which the right of possession and use of the User Product is transferred to the recipient in perpetuity or for a fixed term (regardless of how the transaction is characterized), the Corresponding Source conveyed under this section must be accompanied by the Installation Information. But this requirement does not apply if neither you nor any third party retains the ability to install modified object code on the User Product (for example, the work has been installed in ROM).

The requirement to provide Installation Information does not include a requirement to continue to provide support service, warranty, or updates for a work that has been modified or installed by the recipient, or for the User Product in which it has been modified or installed. Access to a network may be denied when the modification itself materially and adversely affects the operation of the network or violates the rules and protocols for communication across the network.

Corresponding Source conveyed, and Installation Information provided, in accord with this section must be in a format that is publicly documented (and with an implementation available to the public in

source code form), and must require no special password or key for unpacking, reading or copying.

7. Additional Terms.

"Additional permissions" are terms that supplement the terms of this License by making exceptions from one or more of its conditions. Additional permissions that are applicable to the entire Program shall be treated as though they were included in this License, to the extent that they are valid under applicable law. If additional permissions apply only to part of the Program, that part may be used separately under those permissions, but the entire Program remains governed by this License without regard to the additional permissions.

When you convey a copy of a covered work, you may at your option remove any additional permissions from that copy, or from any part of it. (Additional permissions may be written to require their own removal in certain cases when you modify the work.) You may place additional permissions on material, added by you to a covered work, for which you have or can give appropriate copyright permission.

Notwithstanding any other provision of this License, for material you add to a covered work, you may (if authorized by the copyright holders of that material) supplement the terms of this License with terms:

a) Disclaiming warranty or limiting liability differently from the terms of sections 15 and 16 of this License; or

b) Requiring preservation of specified reasonable legal notices or author attributions in that material or in the Appropriate Legal Notices displayed by works containing it; or

c) Prohibiting misrepresentation of the origin of that material, or requiring that modified versions of such material be marked in reasonable ways as different from the original version; or

d) Limiting the use for publicity purposes of names of licensors or authors of the material; or

e) Declining to grant rights under trademark law for use of some trade names, trademarks, or service marks; or

f) Requiring indemnification of licensors and authors of that material by anyone who conveys the material (or modified versions of it) with contractual assumptions of liability to the recipient, for any liability that these contractual assumptions directly impose on

those licensors and authors.

All other non-permissive additional terms are considered "further restrictions" within the meaning of section 10. If the Program as you received it, or any part of it, contains a notice stating that it is governed by this License along with a term that is a further restriction, you may remove that term. If a license document contains a further restriction but permits relicensing or conveying under this License, you may add to a covered work material governed by the terms of that license document, provided that the further restriction does not survive such relicensing or conveying.

If you add terms to a covered work in accord with this section, you must place, in the relevant source files, a statement of the additional terms that apply to those files, or a notice indicating where to find the applicable terms.

Additional terms, permissive or non-permissive, may be stated in the form of a separately written license, or stated as exceptions; the above requirements apply either way.

8. Termination.

You may not propagate or modify a covered work except as expressly provided under this License. Any attempt otherwise to propagate or modify it is void, and will automatically terminate your rights under this License (including any patent licenses granted under the third paragraph of section 11).

However, if you cease all violation of this License, then your license from a particular copyright holder is reinstated (a) provisionally, unless and until the copyright holder explicitly and finally terminates your license, and (b) permanently, if the copyright holder fails to notify you of the violation by some reasonable means prior to 60 days after the cessation.

Moreover, your license from a particular copyright holder is reinstated permanently if the copyright holder notifies you of the violation by some reasonable means, this is the first time you have received notice of violation of this License (for any work) from that copyright holder, and you cure the violation prior to 30 days after your receipt of the notice.

Termination of your rights under this section does not terminate the licenses of parties who have received copies or rights from you under this License. If your rights have been terminated and not permanently

reinstated, you do not qualify to receive new licenses for the same material under section 10.

9. Acceptance Not Required for Having Copies.

You are not required to accept this License in order to receive or run a copy of the Program. Ancillary propagation of a covered work occurring solely as a consequence of using peer-to-peer transmission to receive a copy likewise does not require acceptance. However, nothing other than this License grants you permission to propagate or modify any covered work. These actions infringe copyright if you do not accept this License. Therefore, by modifying or propagating a covered work, you indicate your acceptance of this License to do so.

10. Automatic Licensing of Downstream Recipients.

Each time you convey a covered work, the recipient automatically receives a license from the original licensors, to run, modify and propagate that work, subject to this License. You are not responsible for enforcing compliance by third parties with this License.

An "entity transaction" is a transaction transferring control of an organization, or substantially all assets of one, or subdividing an organization, or merging organizations. If propagation of a covered work results from an entity transaction, each party to that transaction who receives a copy of the work also receives whatever licenses to the work the party’s predecessor in interest had or could give under the previous paragraph, plus a right to possession of the Corresponding Source of the work from the predecessor in interest, if the predecessor has it or can get it with reasonable efforts.

You may not impose any further restrictions on the exercise of the rights granted or affirmed under this License. For example, you may not impose a license fee, royalty, or other charge for exercise of rights granted under this License, and you may not initiate litigation (including a cross-claim or counterclaim in a lawsuit) alleging that any patent claim is infringed by making, using, selling, offering for sale, or importing the Program or any portion of it.

11. Patents.

A "contributor" is a copyright holder who authorizes use under this License of the Program or a work on which the Program is based. The work thus licensed is called the contributor’s "contributor version".

A contributor’s "essential patent claims" are all patent claims

owned or controlled by the contributor, whether already acquired or hereafter acquired, that would be infringed by some manner, permitted by this License, of making, using, or selling its contributor version, but do not include claims that would be infringed only as a consequence of further modification of the contributor version. For purposes of this definition, "control" includes the right to grant patent sublicenses in a manner consistent with the requirements of this License.

Each contributor grants you a non-exclusive, worldwide, royalty-free patent license under the contributor's essential patent claims, to make, use, sell, offer for sale, import and otherwise run, modify and propagate the contents of its contributor version.

In the following three paragraphs, a "patent license" is any express agreement or commitment, however denominated, not to enforce a patent (such as an express permission to practice a patent or covenant not to sue for patent infringement). To "grant" such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party.

If you convey a covered work, knowingly relying on a patent license, and the Corresponding Source of the work is not available for anyone to copy, free of charge and under the terms of this License, through a publicly available network server or other readily accessible means, then you must either (1) cause the Corresponding Source to be so available, or (2) arrange to deprive yourself of the benefit of the patent license for this particular work, or (3) arrange, in a manner consistent with the requirements of this License, to extend the patent license to downstream recipients. "Knowingly relying" means you have actual knowledge that, but for the patent license, your conveying the covered work in a country, or your recipient's use of the covered work in a country, would infringe one or more identifiable patents in that country that you have reason to believe are valid.

If, pursuant to or in connection with a single transaction or arrangement, you convey, or propagate by procuring conveyance of, a covered work, and grant a patent license to some of the parties receiving the covered work authorizing them to use, propagate, modify or convey a specific copy of the covered work, then the patent license you grant is automatically extended to all recipients of the covered work and works based on it.

A patent license is "discriminatory" if it does not include within the scope of its coverage, prohibits the exercise of, or is conditioned on the non-exercise of one or more of the rights that are

specifically granted under this License. You may not convey a covered work if you are a party to an arrangement with a third party that is in the business of distributing software, under which you make payment to the third party based on the extent of your activity of conveying the work, and under which the third party grants, to any of the parties who would receive the covered work from you, a discriminatory patent license (a) in connection with copies of the covered work conveyed by you (or copies made from those copies), or (b) primarily for and in connection with specific products or compilations that contain the covered work, unless you entered into that arrangement, or that patent license was granted, prior to 28 March 2007.

Nothing in this License shall be construed as excluding or limiting any implied license or other defenses to infringement that may otherwise be available to you under applicable patent law.

12. No Surrender of Others' Freedom.

If conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot convey a covered work so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not convey it at all. For example, if you agree to terms that obligate you to collect a royalty for further conveying from those to whom you convey the Program, the only way you could satisfy both those terms and this License would be to refrain entirely from conveying the Program.

13. Use with the GNU Affero General Public License.

Notwithstanding any other provision of this License, you have permission to link or combine any covered work with a work licensed under version 3 of the GNU Affero General Public License into a single combined work, and to convey the resulting work. The terms of this License will continue to apply to the part which is the covered work, but the special requirements of the GNU Affero General Public License, section 13, concerning interaction through a network will apply to the combination as such.

14. Revised Versions of this License.

The Free Software Foundation may publish revised and/or new versions of the GNU General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Program specifies that a certain numbered version of the GNU General Public License "or any later version" applies to it, you have the option of following the terms and conditions either of that numbered version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of the GNU General Public License, you may choose any version ever published by the Free Software Foundation.

If the Program specifies that a proxy can decide which future versions of the GNU General Public License can be used, that proxy's public statement of acceptance of a version permanently authorizes you to choose that version for the Program.

Later license versions may give you additional or different permissions. However, no additional obligations are imposed on any author or copyright holder as a result of your choosing to follow a later version.

15. Disclaimer of Warranty.

THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.

16. Limitation of Liability.

IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MODIFIES AND/OR CONVEYS THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAMAGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

17. Interpretation of Sections 15 and 16.

If the disclaimer of warranty and limitation of liability provided above cannot be given local legal effect according to their terms,

reviewing courts shall apply local law that most closely approximates an absolute waiver of all civil liability in connection with the Program, unless a warranty or assumption of liability accompanies a copy of the Program in return for a fee.

END OF TERMS AND CONDITIONS

How to Apply These Terms to Your New Programs

If you develop a new program, and you want it to be of the greatest possible use to the public, the best way to achieve this is to make it free software which everyone can redistribute and change under these terms.

To do so, attach the following notices to the program. It is safest to attach them to the start of each source file to most effectively state the exclusion of warranty; and each file should have at least the "copyright" line and a pointer to where the full notice is found.

```
    <one line to give the program's name and a brief idea of what it does.>
    Copyright (C) <year>  <name of author>

    This program is free software: you can redistribute it and/or modify
    it under the terms of the GNU General Public License as published by
    the Free Software Foundation, either version 3 of the License, or
    (at your option) any later version.

    This program is distributed in the hope that it will be useful,
    but WITHOUT ANY WARRANTY; without even the implied warranty of
    MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
    GNU General Public License for more details.

    You should have received a copy of the GNU General Public License
    along with this program.  If not, see <https://www.gnu.org/licenses/>.
```

Also add information on how to contact you by electronic and paper mail.

If the program does terminal interaction, make it output a short notice like this when it starts in an interactive mode:

```
    <program>  Copyright (C) <year>  <name of author>
    This program comes with ABSOLUTELY NO WARRANTY; for details type `show w'.
    This is free software, and you are welcome to redistribute it
    under certain conditions; type `show c' for details.
```

The hypothetical commands `show w' and `show c' should show the appropriate parts of the General Public License. Of course, your program's commands

```
might be different; for a GUI interface, you would use an "about box".

  You should also get your employer (if you work as a programmer) or school,
if any, to sign a "copyright disclaimer" for the program, if necessary.
For more information on this, and how to apply and follow the GNU GPL, see
<https://www.gnu.org/licenses/>.

  The GNU General Public License does not permit incorporating your program
into proprietary programs.  If your program is a subroutine library, you
may consider it more useful to permit linking proprietary applications with
the library.  If this is what you want to do, use the GNU Lesser General
Public License instead of this License.  But first, please read
<https://www.gnu.org/licenses/why-not-lgpl.html>.
```

2.49 COPYING.LESSER

```
                   GNU LESSER GENERAL PUBLIC LICENSE
                       Version 3, 29 June 2007

 Copyright (C) 2007 Free Software Foundation, Inc. <https://fsf.org/>
 Everyone is permitted to copy and distribute verbatim copies
 of this license document, but changing it is not allowed.


  This version of the GNU Lesser General Public License incorporates
the terms and conditions of version 3 of the GNU General Public
License, supplemented by the additional permissions listed below.

  0. Additional Definitions.

  As used herein, "this License" refers to version 3 of the GNU Lesser
General Public License, and the "GNU GPL" refers to version 3 of the GNU
General Public License.

  "The Library" refers to a covered work governed by this License,
other than an Application or a Combined Work as defined below.

  An "Application" is any work that makes use of an interface provided
by the Library, but which is not otherwise based on the Library.
Defining a subclass of a class defined by the Library is deemed a mode
of using an interface provided by the Library.

  A "Combined Work" is a work produced by combining or linking an
Application with the Library.  The particular version of the Library
with which the Combined Work was made is also called the "Linked
Version".
```

The "Minimal Corresponding Source" for a Combined Work means the Corresponding Source for the Combined Work, excluding any source code for portions of the Combined Work that, considered in isolation, are based on the Application, and not on the Linked Version.

The "Corresponding Application Code" for a Combined Work means the object code and/or source code for the Application, including any data and utility programs needed for reproducing the Combined Work from the Application, but excluding the System Libraries of the Combined Work.

1. Exception to Section 3 of the GNU GPL.

You may convey a covered work under sections 3 and 4 of this License without being bound by section 3 of the GNU GPL.

2. Conveying Modified Versions.

If you modify a copy of the Library, and, in your modifications, a facility refers to a function or data to be supplied by an Application that uses the facility (other than as an argument passed when the facility is invoked), then you may convey a copy of the modified version:

a) under this License, provided that you make a good faith effort to ensure that, in the event an Application does not supply the function or data, the facility still operates, and performs whatever part of its purpose remains meaningful, or

b) under the GNU GPL, with none of the additional permissions of this License applicable to that copy.

3. Object Code Incorporating Material from Library Header Files.

The object code form of an Application may incorporate material from a header file that is part of the Library. You may convey such object code under terms of your choice, provided that, if the incorporated material is not limited to numerical parameters, data structure layouts and accessors, or small macros, inline functions and templates (ten or fewer lines in length), you do both of the following:

a) Give prominent notice with each copy of the object code that the Library is used in it and that the Library and its use are covered by this License.

b) Accompany the object code with a copy of the GNU GPL and this license

```
    document.

  4. Combined Works.

  You may convey a Combined Work under terms of your choice that,
taken together, effectively do not restrict modification of the
portions of the Library contained in the Combined Work and reverse
engineering for debugging such modifications, if you also do each of
the following:

    a) Give prominent notice with each copy of the Combined Work that
    the Library is used in it and that the Library and its use are
    covered by this License.

    b) Accompany the Combined Work with a copy of the GNU GPL and this license
    document.

    c) For a Combined Work that displays copyright notices during
    execution, include the copyright notice for the Library among
    these notices, as well as a reference directing the user to the
    copies of the GNU GPL and this license document.

    d) Do one of the following:

        0) Convey the Minimal Corresponding Source under the terms of this
        License, and the Corresponding Application Code in a form
        suitable for, and under terms that permit, the user to
        recombine or relink the Application with a modified version of
        the Linked Version to produce a modified Combined Work, in the
        manner specified by section 6 of the GNU GPL for conveying
        Corresponding Source.

        1) Use a suitable shared library mechanism for linking with the
        Library.  A suitable mechanism is one that (a) uses at run time
        a copy of the Library already present on the user's computer
        system, and (b) will operate properly with a modified version
        of the Library that is interface-compatible with the Linked
        Version.

    e) Provide Installation Information, but only if you would otherwise
    be required to provide such information under section 6 of the
    GNU GPL, and only to the extent that such information is
    necessary to install and execute a modified version of the
    Combined Work produced by recombining or relinking the
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
```

```
    the Minimal Corresponding Source and Corresponding Application
    Code. If you use option 4d1, you must provide the Installation
    Information in the manner specified by section 6 of the GNU GPL
    for conveying Corresponding Source.)

  5. Combined Libraries.

  You may place library facilities that are a work based on the
Library side by side in a single library together with other library
facilities that are not Applications and are not covered by this
License, and convey such a combined library under terms of your
choice, if you do both of the following:

    a) Accompany the combined library with a copy of the same work based
    on the Library, uncombined with any other library facilities,
    conveyed under the terms of this License.

    b) Give prominent notice with the combined library that part of it
    is a work based on the Library, and explaining where to find the
    accompanying uncombined form of the same work.

  6. Revised Versions of the GNU Lesser General Public License.

  The Free Software Foundation may publish revised and/or new versions
of the GNU Lesser General Public License from time to time. Such new
versions will be similar in spirit to the present version, but may
differ in detail to address new problems or concerns.

  Each version is given a distinguishing version number. If the
Library as you received it specifies that a certain numbered version
of the GNU Lesser General Public License "or any later version"
applies to it, you have the option of following the terms and
conditions either of that published version or of any later version
published by the Free Software Foundation. If the Library as you
received it does not specify a version number of the GNU Lesser
General Public License, you may choose any version of the GNU Lesser
General Public License ever published by the Free Software Foundation.

  If the Library as you received it specifies that a proxy can decide
whether future versions of the GNU Lesser General Public License shall
apply, that proxy's public statement of acceptance of any version is
permanent authorization for you to choose that version for the
Library.
```

2.50 git_helpers/git_checks.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
```

```
45    declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
46    declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
47    local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
48    LFBFL_send_body_2+=" dès la création du modèle"
49    readonly LFBFL_send_body_2
50    declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
51    declare -r LFBFL_send_body_3=\
52  "Les nouveaux preloadings sont interdits -> #Prefetch()"
53    declare -i LFBFL_i_error=0
54    git diff --cached --name-only\
55      > /tmp/DOSAS_django_git_check1.temp
56    if grep models /tmp/DOSAS_django_git_check1.temp;
57    then
58      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
59      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
60      LFBFL_i_error=1
61    fi
62    if grep serializer /tmp/DOSAS_django_git_check1.temp;
63    then
64      notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65      LFBFL_i_error=1
66    fi
67    rm /tmp/DOSAS_django_git_check1.temp
68    # shellcheck disable=SC2248
69    return ${LFBFL_i_error}
70  }
71
72
73  check_no_abusive_trailing_comma(){
74    # Useful for Python code, although it regularly yields
75    # false positives in Python code
76    # (but it is easy to distinguish by hand).
77    # Use `git commit --no-verify` after check.
78    declare -r LFBFL_send_summary_1="ATTENTION"
79    local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80    LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81    readonly LFBFL_send_body_1
82    declare -i LFBFL_i_error=0
83    git diff --cached -r\
84      > /tmp/DOSAS_django_git_check2.temp
85    if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86    then
87      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88      LFBFL_i_error=1
89    fi
90    rm /tmp/DOSAS_django_git_check2.temp
```

```
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_files_string=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  printf "%s\n" "${LFBFL_files_string}"
  declare -a LFBFL_some_files
  mapfile -t LFBFL_some_files <<< "${LFBFL_files_string}"
  readonly LFBFL_some_files
  declare -i LFBFL_i_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file
  for LFBFL_file in "${LFBFL_some_files[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file}"
    if ! black --check --diff "${LFBFL_file}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

2.51 git_helpers/post-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
```

```
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
```

```
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

2.52 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
```

```
declare -i LFBFL_i_pushd_result2
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  local LFBFL_arg
  local LFBFL_s_format

  declare -i LFBFL_i_log_directory_change=0
  if [[ "$*" == *--log-directory-change* ]]; then
    LFBFL_i_log_directory_change=1
  fi
  readonly LFBFL_i_log_directory_change

  declare -i LFBFL_i_max_comment_line_length=70
```

```
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
      LFBFL_i_max_comment_line_length=$((LFBFL_arg))
      break
    fi
  done
  readonly LFBFL_i_max_comment_line_length

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      "${LFBFL_i_max_comment_line_length}"
  fi
  #-----------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log -p -1\
    | grep '^diff --git' -A 1
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  local LFBFL_split_score_command1
  declare -i LFBFL_i_split_score_command_properties1
  get_split_score_simple 1 "${LFBFL_i_max_comment_line_length}" _
  LFBFL_split_score_command1="${get_split_score_result}"
  LFBFL_i_split_score_command_properties1="${i_get_split_score_result2}"
  readonly LFBFL_split_score_command1
  readonly LFBFL_i_split_score_command_properties1
  local LFBFL_split_score_command2
```

```
131     declare -i LFBFL_i_split_score_command_properties2
132     get_split_score_simple 1 "${LFBFL_i_max_comment_line_length}" /
133     LFBFL_split_score_command2="${get_split_score_result}"
134     LFBFL_i_split_score_command_properties2="${i_get_split_score_result2}"
135     readonly LFBFL_split_score_command2
136     readonly LFBFL_i_split_score_command_properties2
137     declare -i LFBFL_i_renaming_happened=0
138     declare -i LFBFL_i
139     declare -i LFBFL_j
140     # declare -r LFBFL_special_file_name="files_names_listing.txt"
141     declare -r LFBFL_sfn="files_names_listing.txt"
142     local LFBFL_diff_line
143     local LFBFL_similarity_line
144     local LFBFL_old_file_path
145     local LFBFL_old_file_name
146     local LFBFL_old_file_directory
147     local LFBFL_new_file_path
148     local LFBFL_new_file_name
149     local LFBFL_new_file_directory
150     local LFBFL_useful_file_name
151     local LFBFL_extension
152     local LFBFL_comment_prefix
153     local LFBFL_comment_prefix2
154     local LFBFL_old_file_name2
155     local LFBFL_old_file_directory2
156     local LFBFL_new_file_name2
157     local LFBFL_new_file_directory2
158     local LFBFL_new_comment
159     declare -i LFBFL_i_copyright_line_number
160     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
161     declare -i LFBFL_i_line_count
162     declare -i LFBFL_i_lines_after
163     local LFBFL_temp_file_path
164     for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
165       LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
166       LFBFL_i=$((LFBFL_i + 1))
167       LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
168       LFBFL_i=$((LFBFL_i + 1))
169       # We need to increment 3 times since grep adds "--" lines between.
170       LFBFL_i=$((LFBFL_i + 1))
171       if [[ LFBFL_i_verbose -eq 1 ]]; then
172         printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
173           "${LFBFL_diff_line}"\
174           "${LFBFL_similarity_line}"
175       fi
176       if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
```

```
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. ------------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
```

```
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_i_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_i_split_score_command_properties1}"
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_i_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
```

```
      "${LFBFL_i_split_score_command_properties2}"
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_i_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_i_split_score_command_properties1}"
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_i_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_i_split_score_command_properties2}"
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. ----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
      ]];
    then
      if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
      then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
```

```
315         LFBFL_new_comment+="${LFBFL_comment_prefix}to"
316         LFBFL_new_comment+=$'\n'
317         LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
318       fi
319       if [[ LFBFL_i_verbose -eq 1 ]]; then
320         printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
321       fi
322       # Find line with ©Copyright. -------------------------------------
323       LFBFL_i_copyright_line_number=$(
324         grep -n '©Copyright' "${LFBFL_new_file_path}"\
325         | head --lines=1\
326         | cut -d ':' -f 1
327       )
328       if [[ LFBFL_i_verbose -eq 1 ]]; then
329         printf "LFBFL_i_copyright_line_number: %s\n"\
330           "${LFBFL_i_copyright_line_number}"
331       fi
332       if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
333         notify-send "${LFBFL_send_summary_2}"\
334           "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
335         if [[ LFBFL_i_verbose -eq 1 ]]; then
336           printf "No copyright line. Skipping file.\n"
337         fi
338         continue
339       fi
340       # Get total number of lines. -------------------------------------
341       LFBFL_i_line_count=$(
342         wc -l\
343         < "${LFBFL_new_file_path}"
344       )
345       if [[ LFBFL_i_verbose -eq 1 ]]; then
346         printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
347       fi
348       # Compute number of lines after. ---------------------------------
349       LFBFL_i_lines_after=$((
350         LFBFL_i_line_count - LFBFL_i_copyright_line_number
351       ))
352       if [[ LFBFL_i_verbose -eq 1 ]]; then
353         printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
354       fi
355       # Update file. ---------------------------------------------------
356       LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
357       {
358         head --lines="${LFBFL_i_copyright_line_number}"\
359           "${LFBFL_new_file_path}"
360         printf "%s\n" "${LFBFL_new_comment}"
```

```
      tail --lines="${LFBFL_i_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. ---------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit ---------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    git commit --message="${LFBFL_message}"
  fi
}
```

2.53 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
```

```
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

2.54 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
```

```
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don't see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don't need "pass" XD.
    """


def IMAP_transfert(
```

```
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
```

```
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
```

```
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
```

```
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-------------------------------------------------------\n"
```

```
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

2.55 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
-----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
```

```
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

2.56 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
```

```
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

2.57 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

2.58 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
[![Security:bandit:badge]][Bandit-GitHub.com]
[![Code-style:black:badge]][Black-GitHub.com]
[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.
```

```
I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.

But the files "pre-commit" and "post-commit" were not renamed,
since it is not possible to give them other names to use them as
pre-commit/post-commit hooks in `git`.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts
```

```
[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

[GitHub-top-language-badge]: https://img.shields.io/github/\
languages/top/llyaudet/DevOrSysAdminScripts

[GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

2.59 unicode.libr.php

```
<?php
/*
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  $i_code_point_in_decimal_notation
){
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * (2**3)){
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % (2**6)
    );
    $i_first_byte_significant_bits = intdiv(
      (
```

```
45          $i_code_point_in_decimal_notation
46          - $i_last_byte_significant_bits
47        ),
48        2**6,
49      );
50      $s_result = chr(
51        $i_first_byte_base_value
52        + $i_first_byte_significant_bits
53      );
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 256 * 256){
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
79        // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
80        $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
81        $i_rest = intdiv(
82          $i_rest - $i_significant_bits,
83          2**$arr_data["i_bits"],
84        );
85        $s_result = (  // .= or =.
86          chr($i_significant_bits + $arr_data["i_base_value"])
87          .$s_result
88        );
89      }
90      return $s_result;
```

```
91    }
92    // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
93    if($i_code_point_in_decimal_notation < 256 * 256 * (2**5)){
94      $i_first_byte_base_value = 240;
95      $arr_arr_data_per_byte_reverse = [
96        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
97        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
98        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
99        ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
100     ];
101     $i_rest = $i_code_point_in_decimal_notation;
102     $s_result = "";
103     foreach($arr_arr_data_per_byte_reverse as $arr_data){
104       // _Pragma unroll
105       $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
106       $i_rest = intdiv(
107         $i_rest - $i_significant_bits,
108         2**$arr_data["i_bits"],
109       );
110       $s_result = (  // .= or =.
111         chr($i_significant_bits + $arr_data["i_base_value"])
112         .$s_result
113       );
114     }
115     return $s_result;
116   }
117   throw new Exception(
118     "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
119     ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
120     ." 3 octets au lieu de 4 :face-palm:."
121     ." Du coup, ça peut donner quelques \"surprises\"."
122     ." Sinon, y'a aussi ASCII dans MySQL."
123     ." Et comme ça joue sur la mémoire et les performances,"
124     ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
125     ." :)"
126   );
127 }
128
129
130
131 function hexa_code_point_to_UTF8(
132   $s_code_point_in_hexadecimal_notation
133 ){
134   $s_code_point_in_hexadecimal_notation = strtoupper(
135     $s_code_point_in_hexadecimal_notation
136   );
```

```
137     $i_code_point_in_decimal_notation = 0;
138     $i_ord_0 = ord('0');
139     $i_ord_A = ord('A');
140     $i_digit_offset = -$i_ord_0;
141     $i_letter_offset = 10 - $i_ord_A;
142     for(
143       $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144       $i < $i_max;
145       ++$i
146     ){
147       $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
148       $i_code_point_in_decimal_notation *= 16;
149       if(ctype_digit($s_hexa_digit)){
150         $i_code_point_in_decimal_notation += (
151           ord($s_hexa_digit) + $i_digit_offset
152         );
153         continue;
154       }
155       if(ctype_xdigit($s_hexa_digit)){
156         $i_code_point_in_decimal_notation += (
157           ord($s_hexa_digit) + $i_letter_offset
158         );
159         continue;
160       }
161       throw new Exception("Not an hexadecimal digit.");
162     }
163     return decimal_code_point_to_UTF8(
164       $i_code_point_in_decimal_notation
165     );
166   }
167
168
169
170   function check_string_is_valid_ASCII($s_string){
171     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
172       if(ord($s_string[$i]) > 127){
173         throw new Exception(
174           "Non-ASCII character found at octet position "
175           .$i
176           ., value "
177           .ord($s_string[$i])
178           ."."
179         );
180       }
181     }
182     return true;
```

```
183 }
184
185
186
187 function check_file_is_valid_ASCII($s_file_name){
188   $s_string = file_get_contents($s_file_name);
189   if($s_string === false){
190     throw new Exception("File ".$s_file_name." not found.");
191   }
192   return check_string_is_valid_ASCII($s_string);
193 }
194
195
196
197 function check_string_is_valid_UTF8($s_string){
198   $i_continuation_octet_needed = 0;
199   $i_offset_in_octets_from_string_start = 0;
200   $i_offset_in_characters_from_string_start = -1;
201   $i_character_start_position_from_string_start = 0;
202   $i_current_line_number = 1;
203   $i_offset_in_octets_from_line_start = -1;
204   $i_offset_in_characters_from_line_start = -1;
205   $i_character_start_position_from_line_start = -1;
206   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
207     $i_current_octet = ord($s_string[$i]);
208     $i_offset_in_octets_from_string_start = $i;
209     $i_offset_in_octets_from_line_start += 1;
210     $i_character_start_position_from_line_start += 1;
211     if($i_continuation_octet_needed > 0){
212       if($i_current_octet < 128 || $i_current_octet >= 192){
213         throw new Exception(
214           "Non-UTF8 character found on line "
215           .$i_current_line_number
216           ."; the octet "
217           .($i_offset_in_octets_from_line_start + 1)
218           ", part of the character "
219           .($i_offset_in_characters_from_line_start + 1)
220           ", has value "
221           .$i_current_octet
222           ." which is not a continuation octet."
223           ." This character starts at octet "
224           .($i_character_start_position_from_line_start + 1)
225           ." of the current line."
226           ." (Sequential positions without line splitting:"
227           ." This is at character "
228           .($i_offset_in_characters_from_string_start + 1)
```

```
229           ." and octet "
230           .($i_offset_in_octets_from_string_start + 1)
231           "."
232           ." This character starts at octet "
233           .($i_character_start_position_from_string_start + 1)
234           ".)",
235         );
236       }
237       --$i_continuation_octet_needed;
238       continue;
239     }
240
241     $i_character_start_position_from_string_start = $i;
242     $i_offset_in_characters_from_line_start += 1;
243     $i_offset_in_characters_from_string_start += 1;
244     if($i_current_octet < 128){ // 0xxxxxxx ASCII
245       if($s_string[$i] === "\n"){
246         $i_current_line_number += 1;
247         $i_offset_in_octets_from_line_start = -1;
248         $i_offset_in_characters_from_line_start = -1;
249         $i_character_start_position_from_line_start = -1;
250       }
251       continue;
252     }
253     if($i_current_octet >= 128 && $i_current_octet < 192){
254       throw new Exception(
255         "Non-UTF8 character found on line "
256         .$i_current_line_number
257         ."; the octet "
258         .($i_offset_in_octets_from_line_start + 1)
259         ", part of the character "
260         .($i_offset_in_characters_from_line_start + 1)
261         ", has value "
262         .$i_current_octet
263         ." which is a continuation octet."
264         ." This character starts at octet "
265         .($i_character_start_position_from_line_start + 1)
266         ." of the current line."
267         ." (Sequential positions without line splitting:"
268         ." This is at character "
269         .($i_offset_in_characters_from_string_start + 1)
270         ." and octet "
271         .($i_offset_in_octets_from_string_start + 1)
272         "."
273         ." This character starts at octet "
274         .($i_character_start_position_from_string_start + 1)
```

```
275          ".)",
276        );
277      }
278      if($i_current_octet >= 192 && $i_current_octet < 224){
279        // 110xxxxx 10xxxxxx
280        $i_continuation_octet_needed = 1;
281        continue;
282      }
283      if($i_current_octet >= 224 && $i_current_octet < 240){
284        // 1110xxxx 10xxxxxx 10xxxxxx
285        $i_continuation_octet_needed = 2;
286        continue;
287      }
288      if($i_current_octet >= 240 && $i_current_octet < 248){
289        // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
290        $i_continuation_octet_needed = 3;
291        continue;
292      }
293      throw new Exception(
294        "Non-UTF8 character found on line "
295        .$i_current_line_number
296        ."; the octet "
297        .($i_offset_in_octets_from_line_start + 1)
298        ", part of the character "
299        .($i_offset_in_characters_from_line_start + 1)
300        ", has value "
301        .$i_current_octet
302        ." which is invalid."
303        ." This character starts at octet "
304        .($i_character_start_position_from_line_start + 1)
305        ." of the current line."
306        ." (Sequential positions without line splitting:"
307        ." This is at character "
308        .($i_offset_in_characters_from_string_start + 1)
309        ." and octet "
310        .($i_offset_in_octets_from_string_start + 1)
311        "."
312        ." This character starts at octet "
313        .($i_character_start_position_from_string_start + 1)
314        ".)",
315      );
316    }
317    return true;
318  }
319
320
```

```
321
322 function check_file_is_valid_UTF8($s_file_name){
323   $s_string = file_get_contents($s_file_name);
324   if($s_string === false){
325     throw new Exception("File ".$s_file_name." not found.");
326   }
327   return check_string_is_valid_UTF8($s_string);
328 }
329
330
331
332 /*
333 ?>
334 <?php
335 require_once("./unicode.libr.php");
336 var_dump(hexa_code_point_to_UTF8("002B"));
337 var_dump(hexa_code_point_to_UTF8("00E6"));
338 var_dump(hexa_code_point_to_UTF8("1400"));
339 var_dump(hexa_code_point_to_UTF8("10111"));
340 // https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
341 // Some invisible unicode characters that are a problem:
342 // 2000-200F 2028-202F 205F 2060-206F
343 $arr = [
344   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
345   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
346   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
347   '205F',
348   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
349   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
350 ];
351 $arr_s = [];
352 $arr_s2 = [];
353 $arr_s3 = [];
354 foreach($arr as $s_hexa){
355   $s = hexa_code_point_to_UTF8($s_hexa);
356   $arr_s []= $s;
357   if(strlen($s) === 2){
358     $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
359   }
360   if(strlen($s) === 3){
361     $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
362       .'\x'.bin2hex($s[2]);
363   }
364   $arr_s2 []= $s2;
365   $s3 = "  | sed -e 's/".$s2."/ /g'\\";
366   $arr_s3 []= $s3;
```

```
  echo($s3."\n");
}
//*/

```

2.60 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
```

```
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

2.61 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1 file_path
  # $2 download_URL
  # $3 correct_sha512
  # $4 optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
```

```
  readonly LFBFL_verbose
  if [[ ! -f "$1" ]]; then
    wget "${LFBFL_verbose}" -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut -f1 -d' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    return 1
  fi
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !